Advanced Topics

Recursive and Convolutional NNs,
Domain Adaptation, Multitask, Multimodal,
GAN’s and Reinforcement Learning

In this lecture

- Convolutional neural network
- Recursive neural network
- Domain adaptation
- Multi-task learning
- Multi-source, multi-target
- Multi-modal learning
- Generative adversarial neural network
- Reinforcement learning

Convolutional Neural Networks

Convolutional Neural Networks

“ConvNet” architecture originally proposed for images

Convolutional Neural Networks

"ConvNet" architecture originally proposed for images

The assumption was that spatially local information is very important in images

Convolutional Neural Networks

"ConvNet" architecture originally proposed for images

The assumption was that spatially local information is very important in images

Objects are "groups" of pixels that are nearby

Convolutional Neural Networks

ConvNet's propose the concept of a *filter* (also known as a *kernel*) - which acts like a **feature detector**

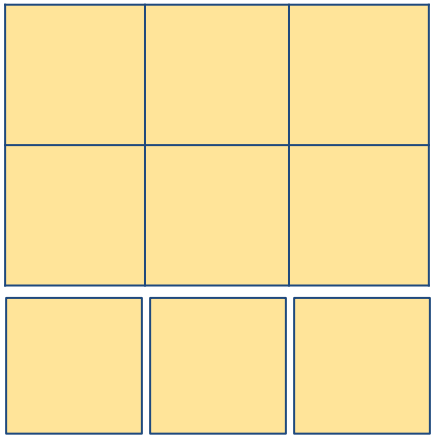
Convolutional Neural Networks

ConvNet's propose the concept of a *filter* (also known as a *kernel*) - which acts like a **feature detector**

It does the same job as a *neuron* in the networks we've seen so far - each *filter* considers some aspect of the input and its outputs are a measure of how much the filter supports the particular aspect/feature

Convolutional Neural Networks

Filters are defined by some width and height, and are usually square



We slide a filter over the entire image, and therefore produce a corresponding second image from the activations of the filter

Convolutional Neural Networks

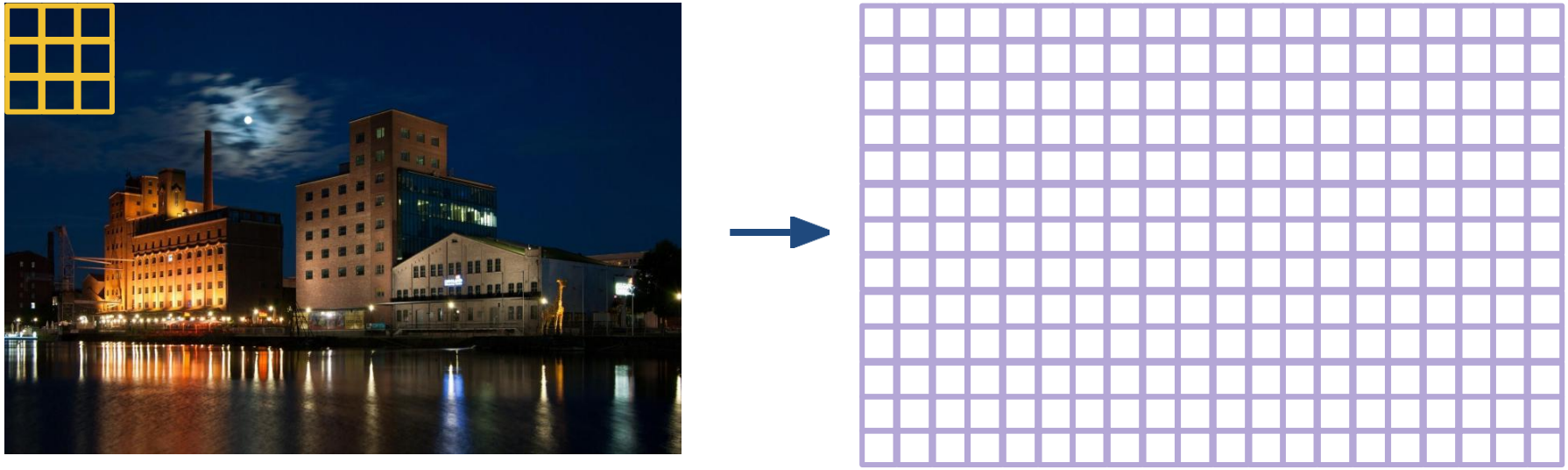
Let's look at a concrete example
Consider an image of size 15x22 pixels

22

15

Convolutional Neural Networks

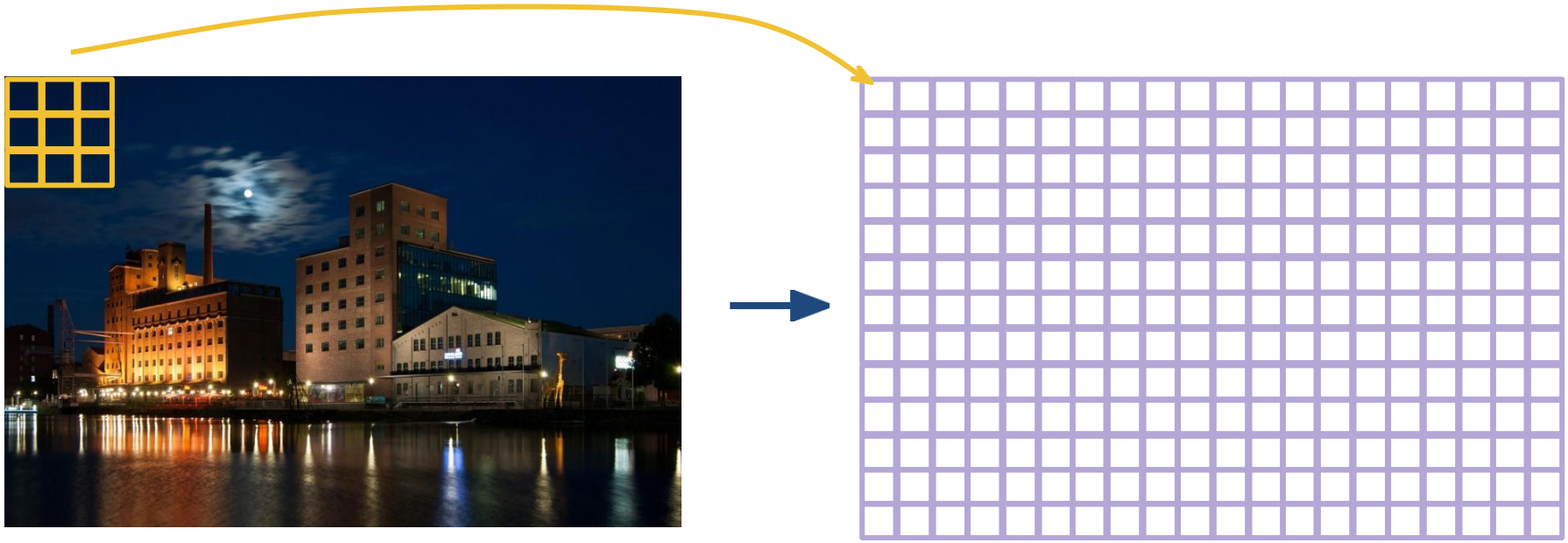
Consider a small 3x3 filter that has learned to detect buildings



"Activation map"
Generated Image

Convolutional Neural Networks

The filter will look at the 9 pixels underneath it,
and output **one** value



"Activation map"
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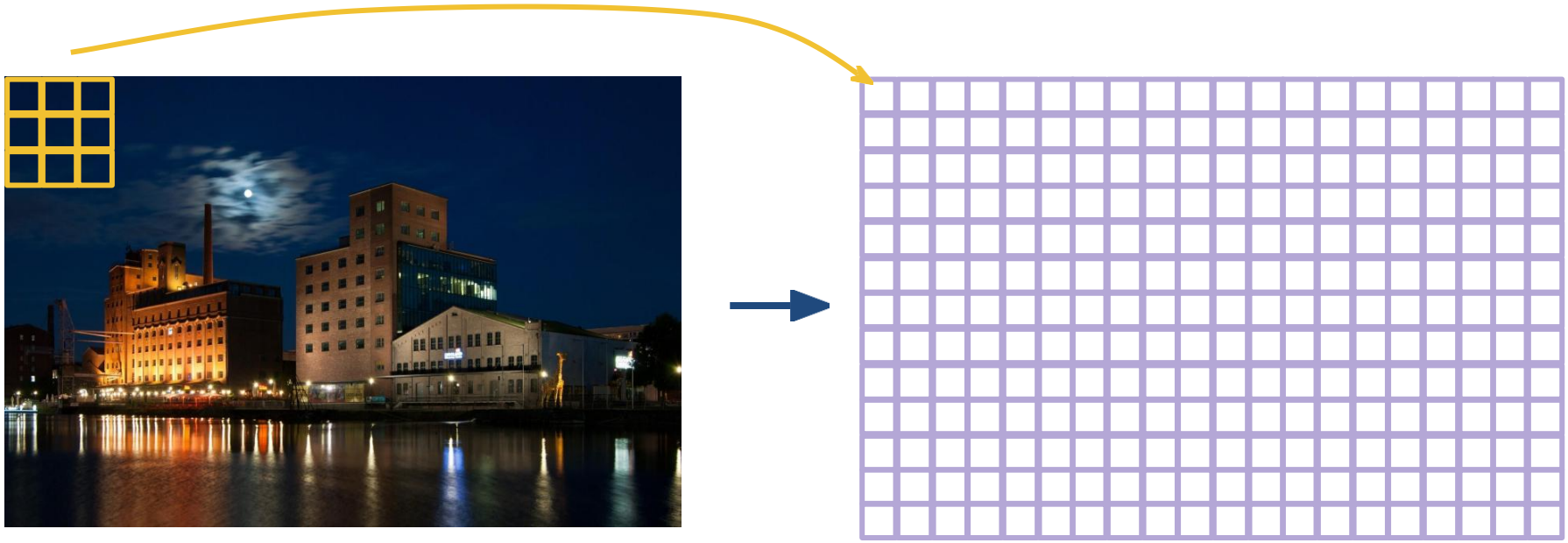
Convolutional Neural Networks

This value can be considered as its “support” for the feature it is learning, in this case “buildings”



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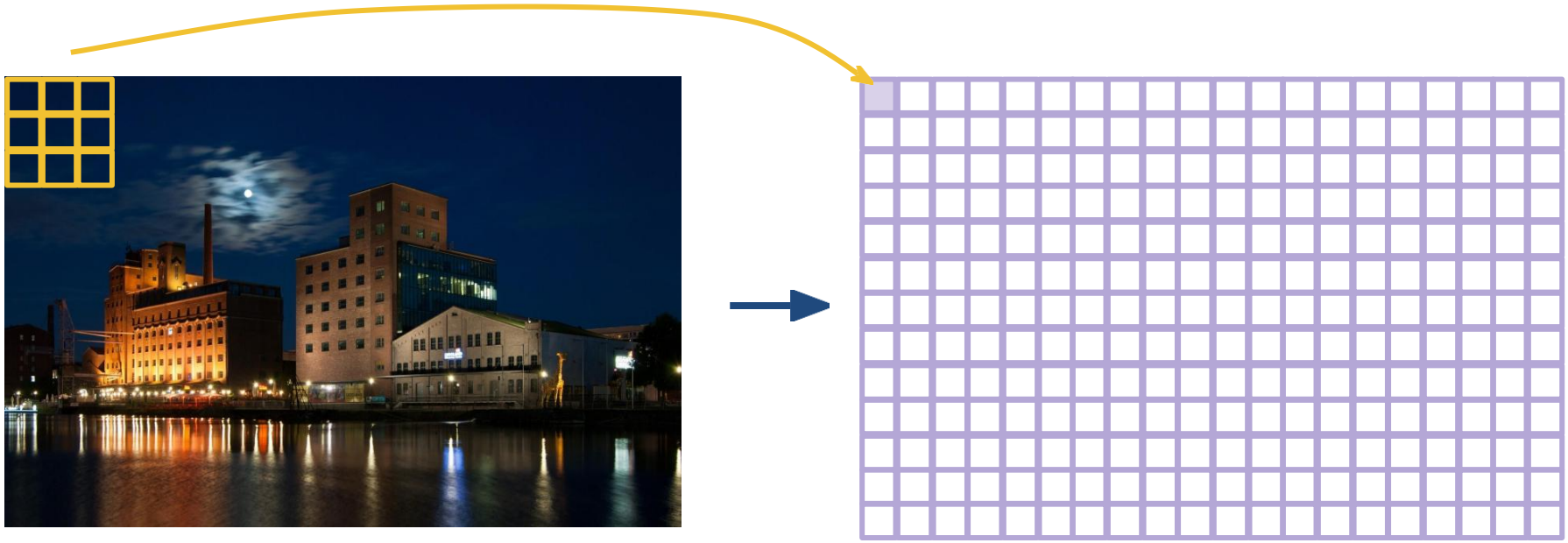
Convolutional Neural Networks

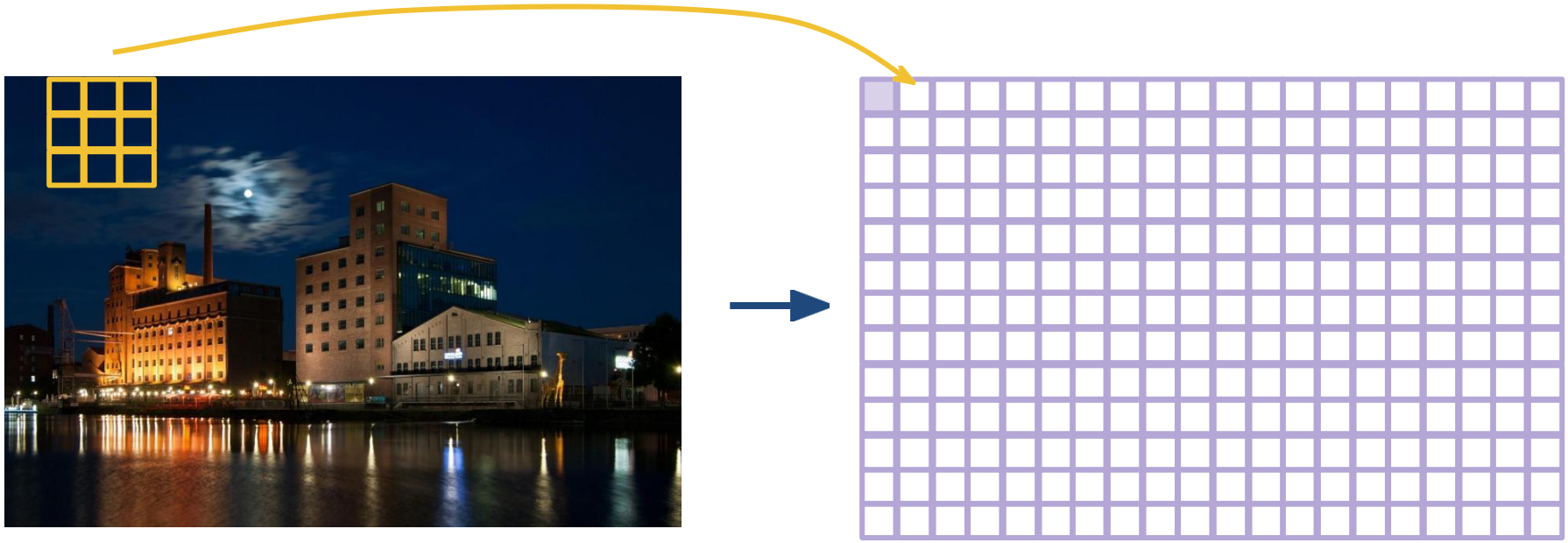
Since there is no building underneath the filter, it will output a low value in this case



"Activation map"
Generated Image

Convolutional Neural Networks

We then slide the window **by a pixel** to the right,
and repeat the process



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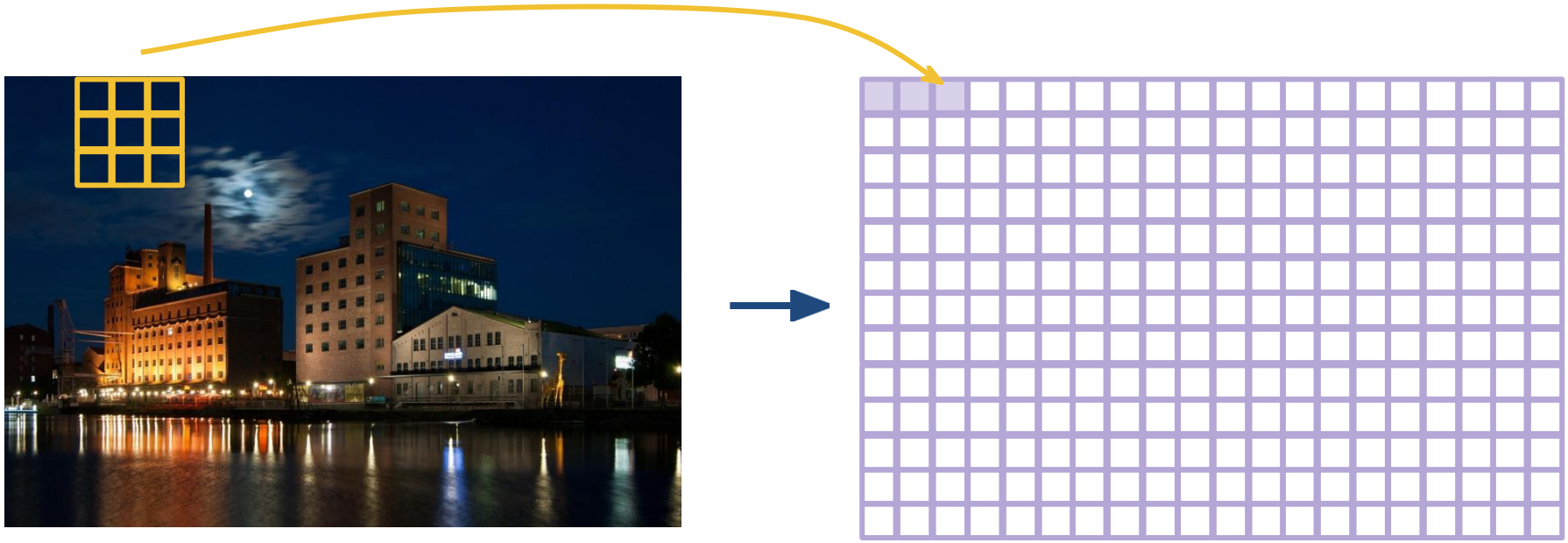

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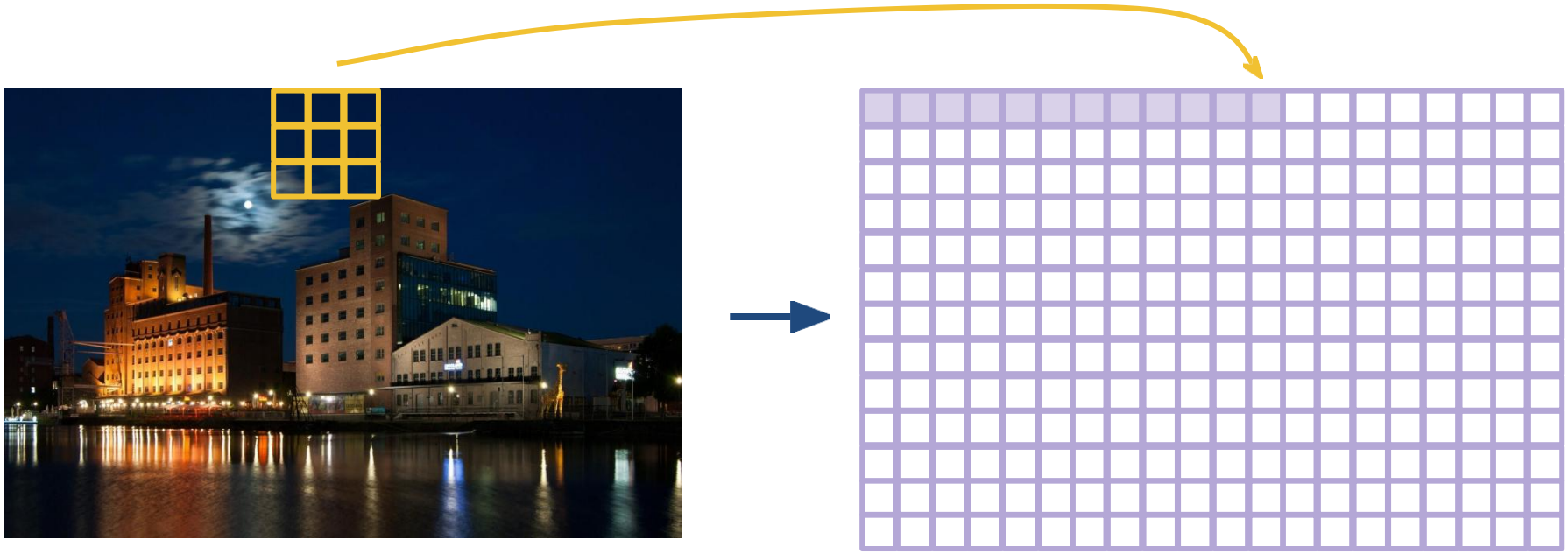
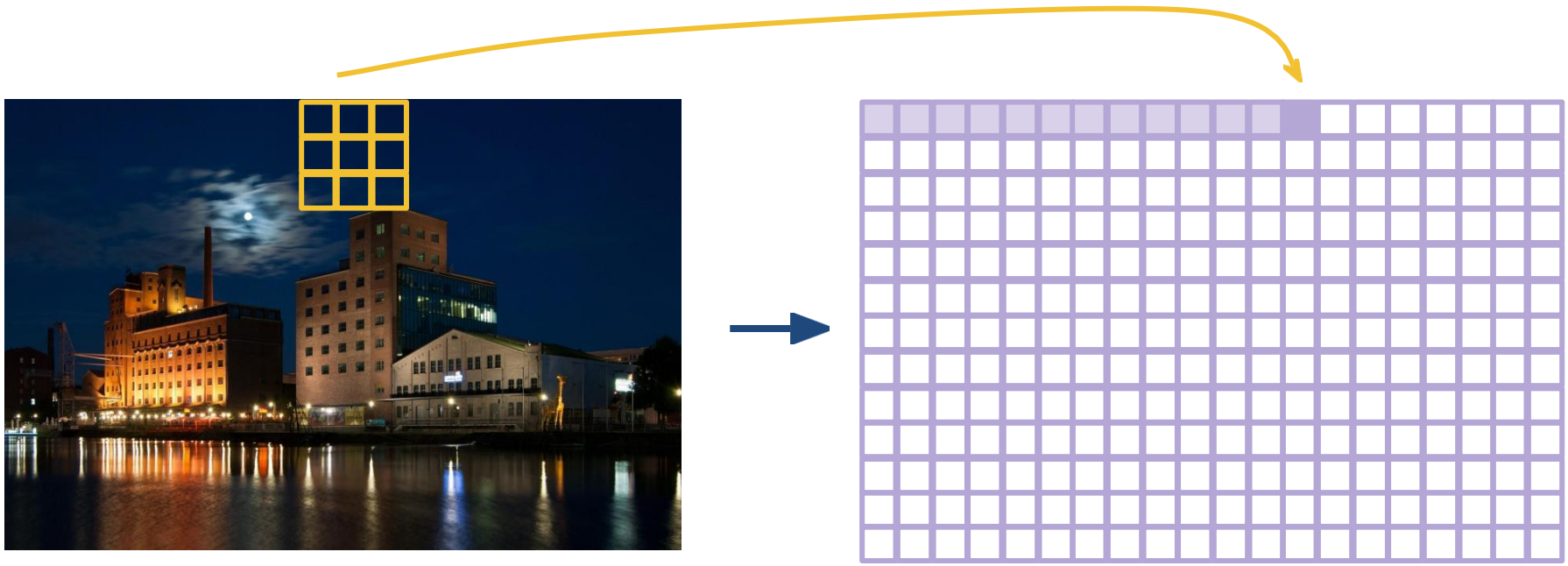


"Activation map"
Generated Image

Convolutional Neural Networks

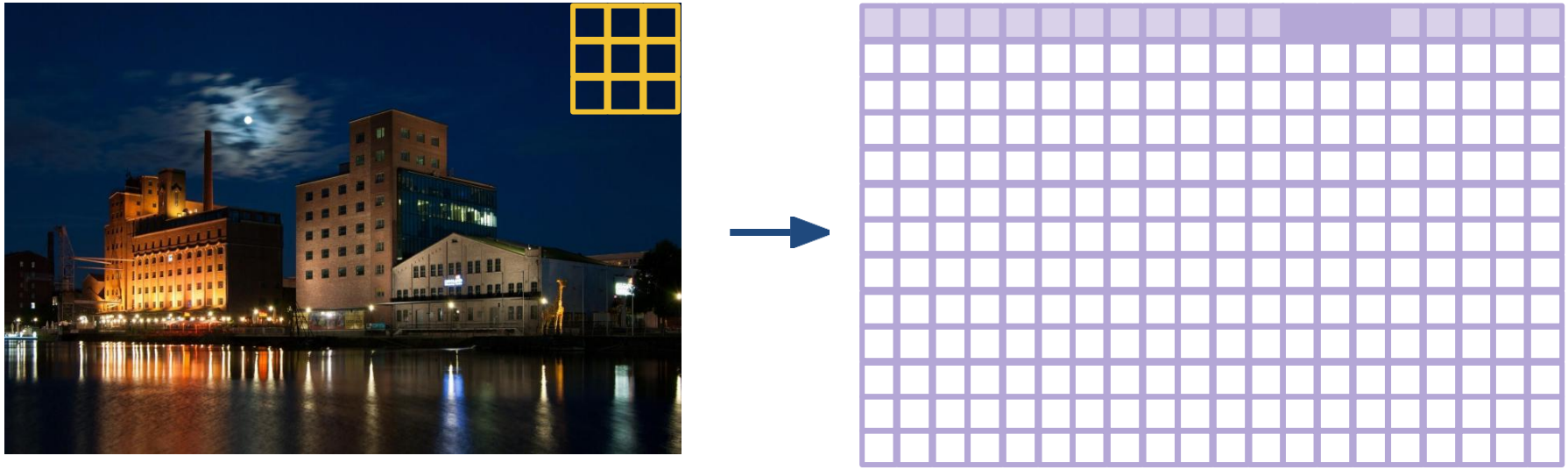
In this case, the filter finally sees some portion of some building, so its output will be slightly higher



"Activation map"
Generated Image

Convolutional Neural Networks

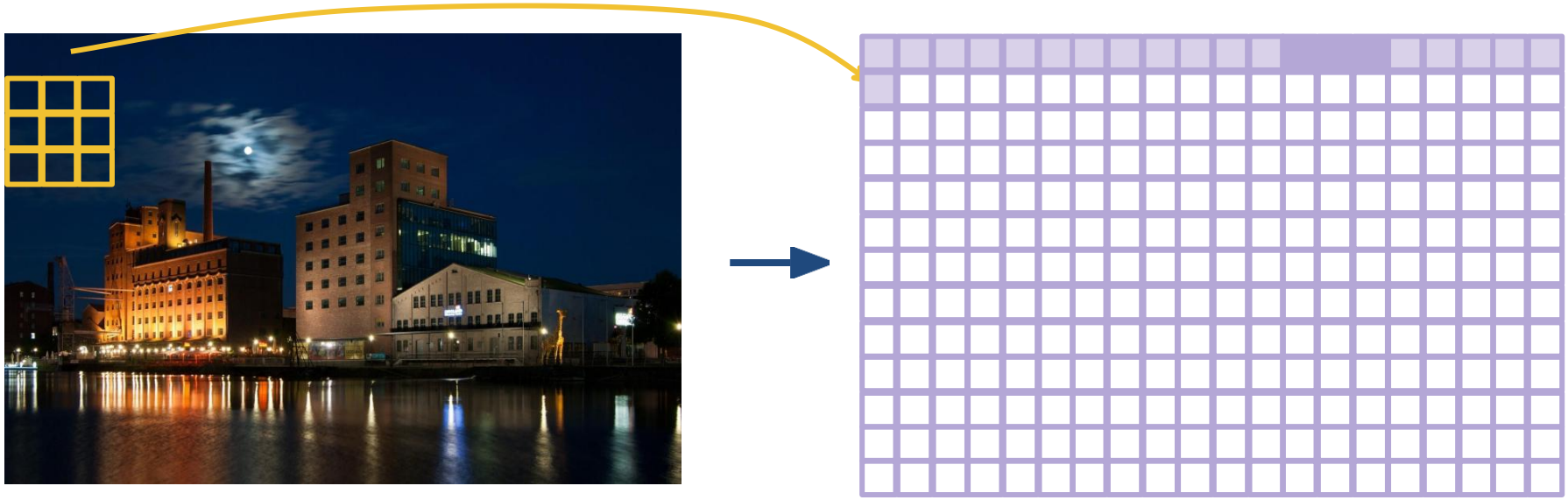
We repeat the process until we reach the end of the row



"Activation map"
Generated Image

Convolutional Neural Networks

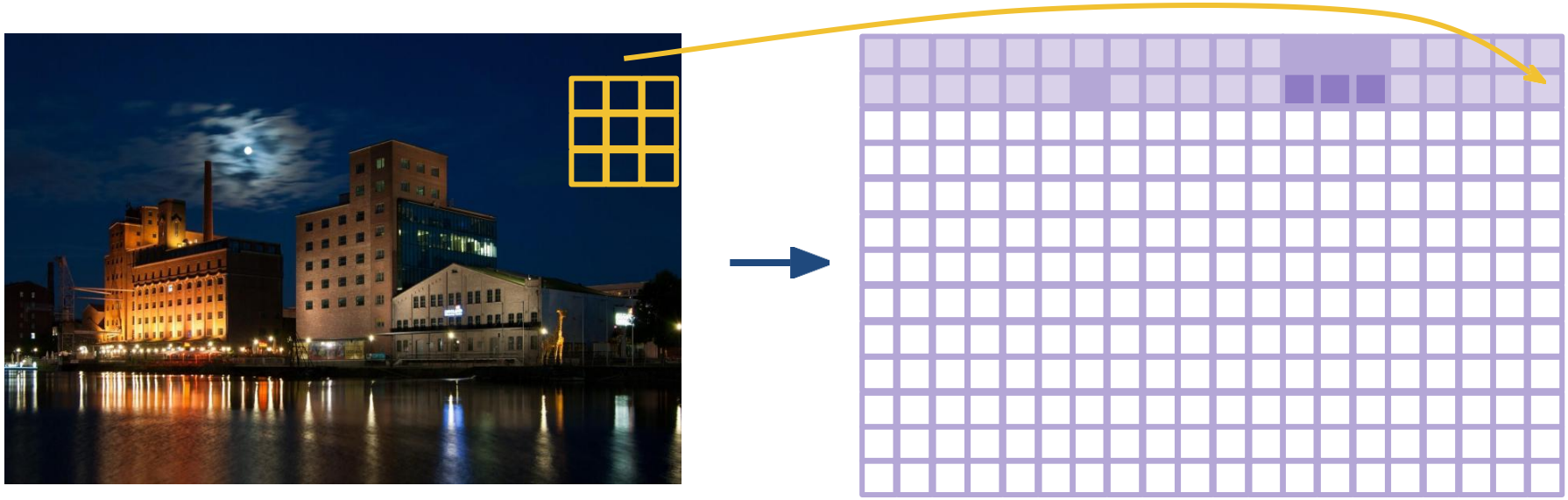
We shift **one pixel down** and repeat the entire process, filling in the second row of the "activation map"



"Activation map"
Generated Image

Convolutional Neural Networks

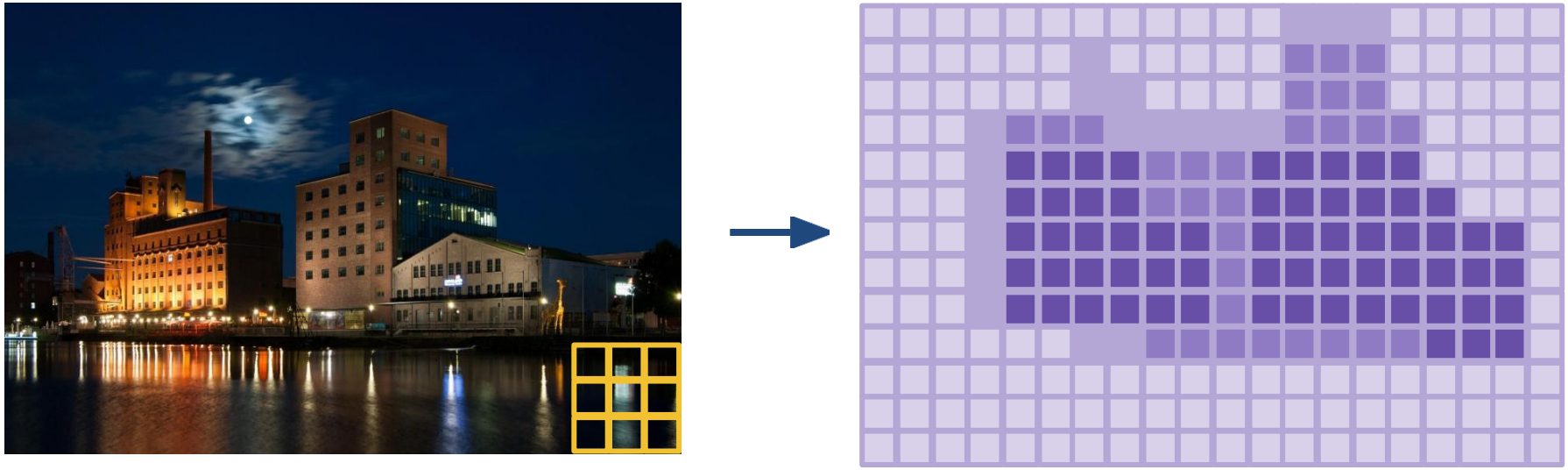
We continue until the end of the row



“Activation map”
Generated Image

Convolutional Neural Networks

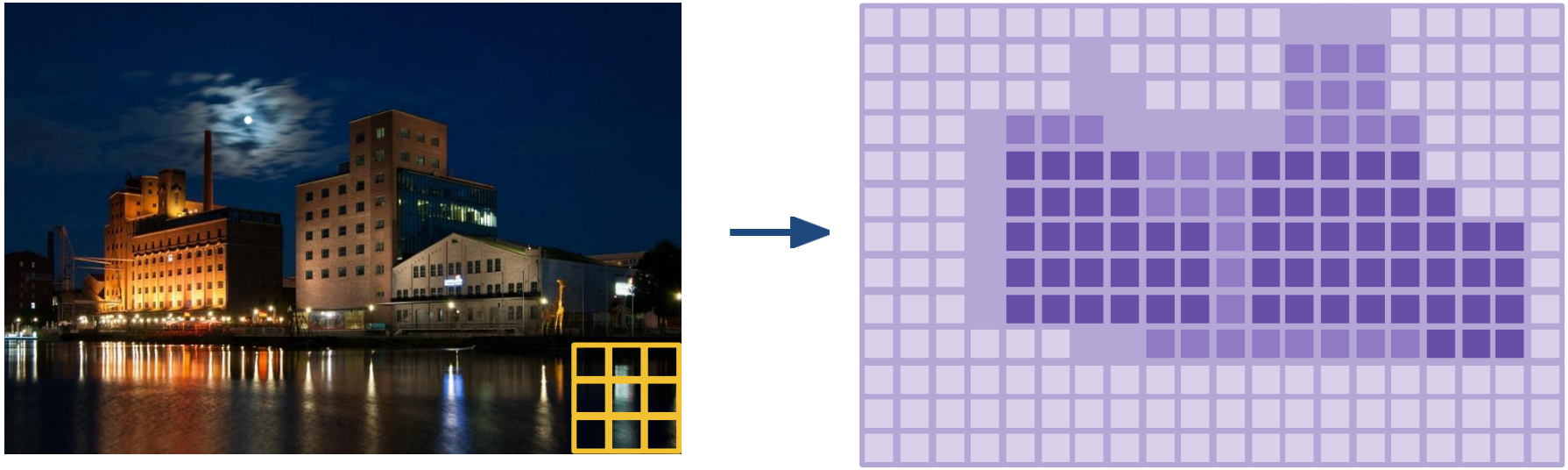
And repeat the entire process for the whole image



"Activation map"
Generated Image

Convolutional Neural Networks

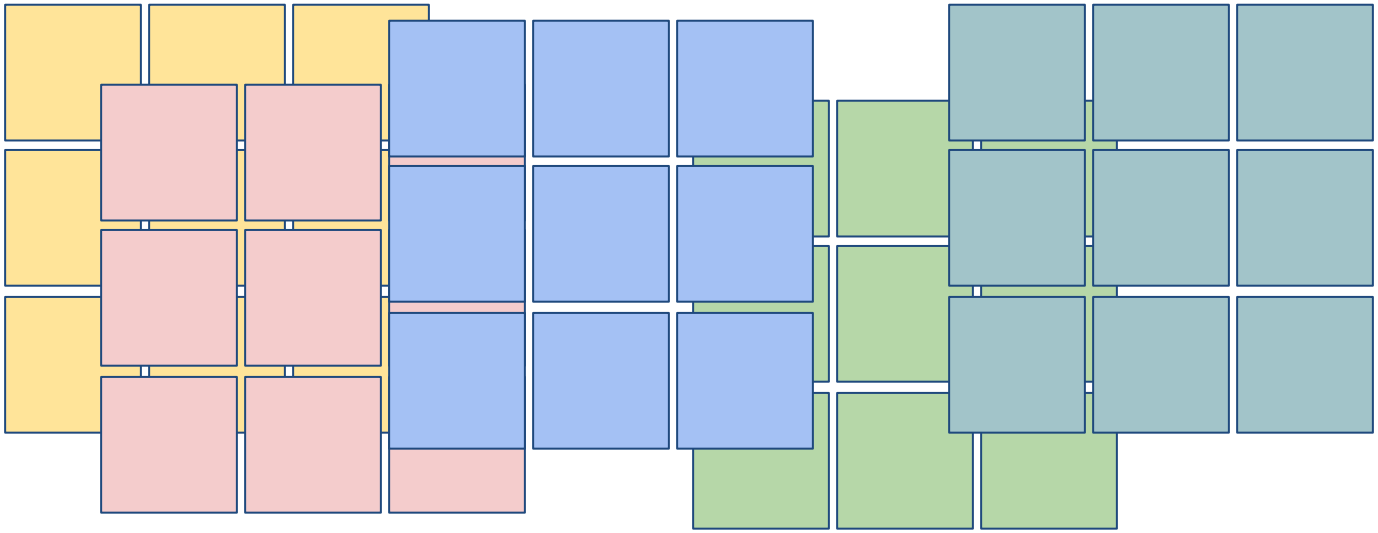
The higher layers can now use this activation map to create richer features based on “buildings information”



“Activation map”
Generated Image

Convolutional Neural Networks

Usually, we have a lot of filters, so each of them can learn different features from the input image



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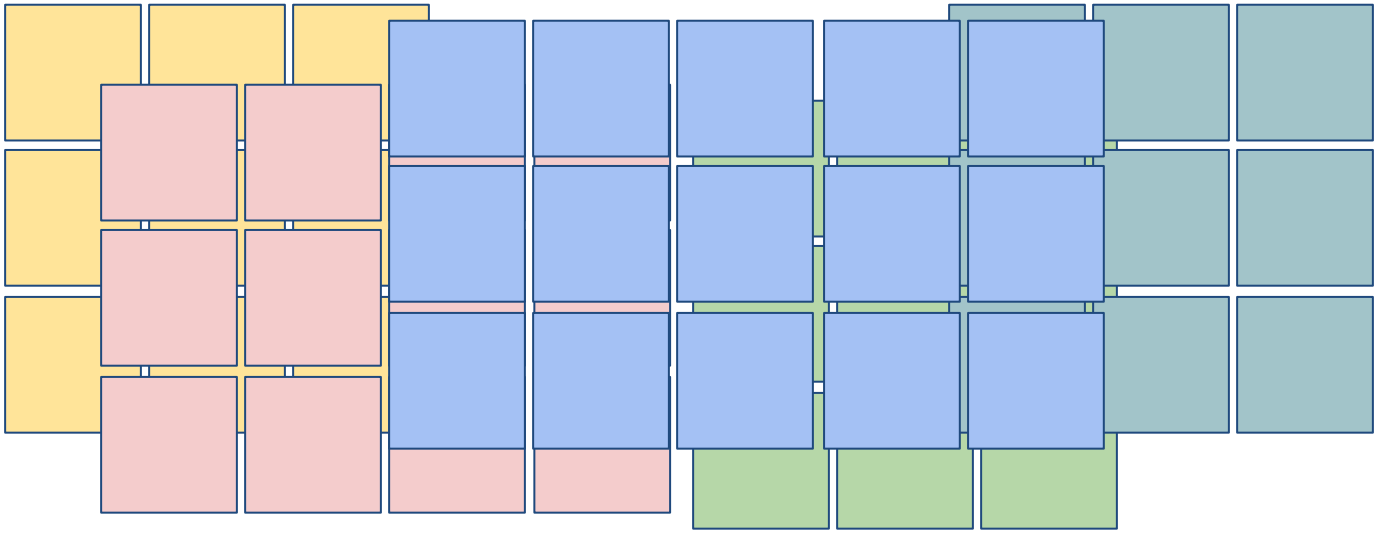

Although not common in images, for text, we can also have filters of different sizes!

Convolutional Neural Networks

Some terminologies:

- **Stride:** The number of pixels to shift at each step
- **Depth:** The depth of a Conv layer is the number of filters in it - If we have *N* filters, we will have *N* activation maps

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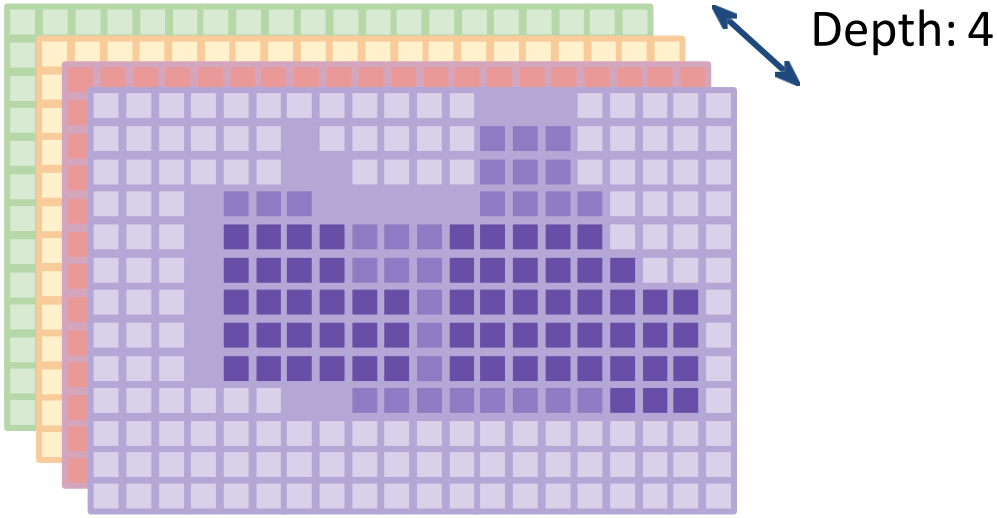

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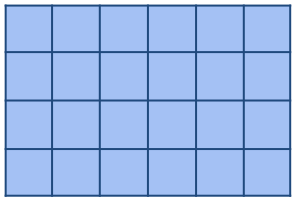

4x6 image

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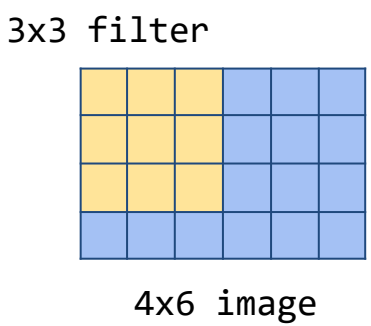

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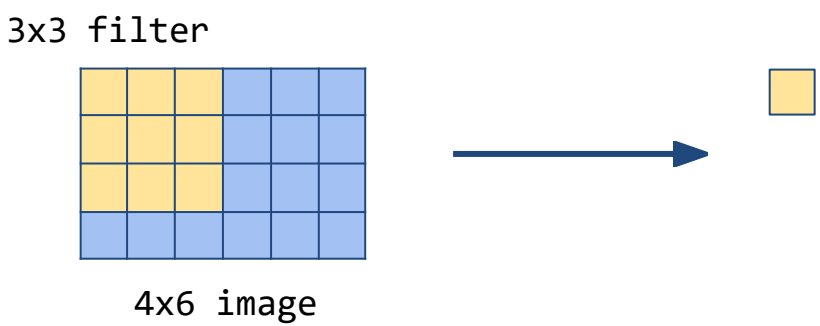

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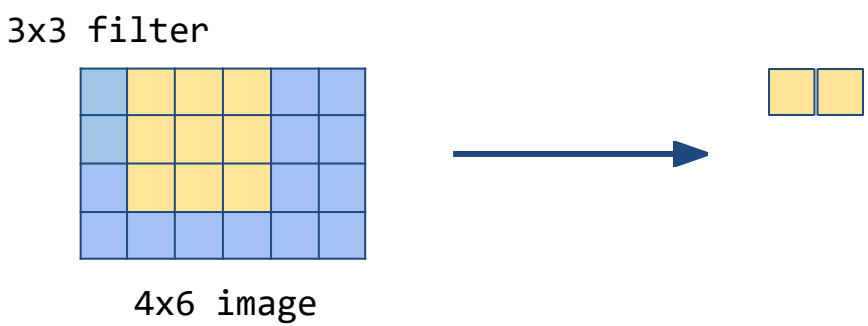

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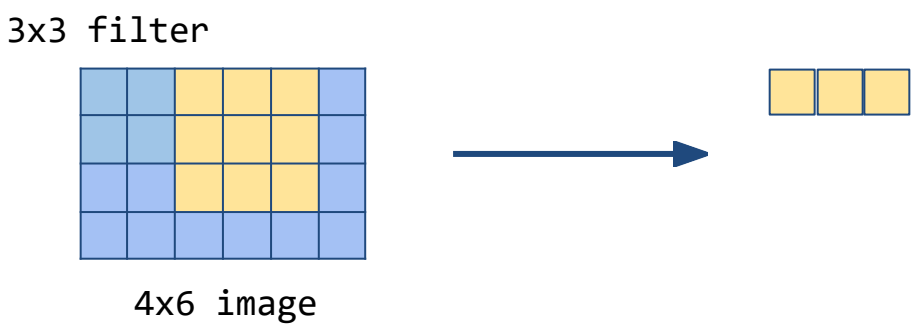

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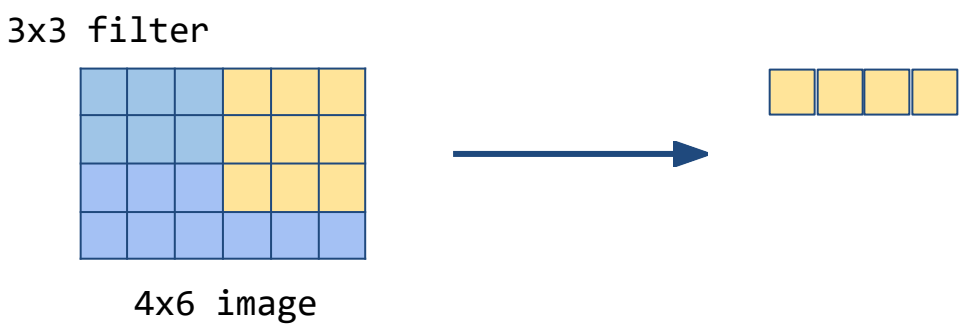

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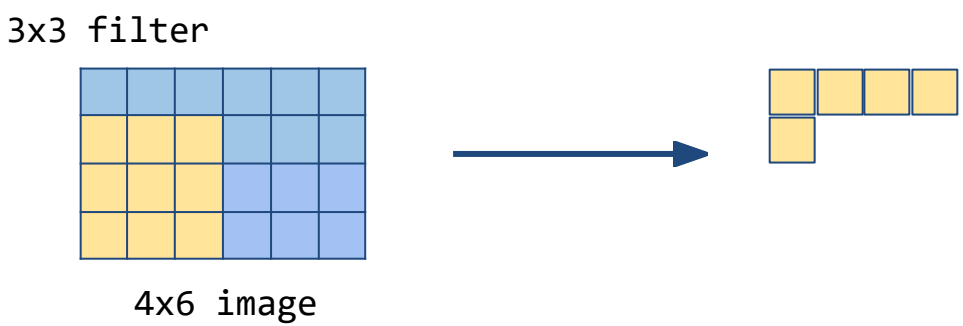

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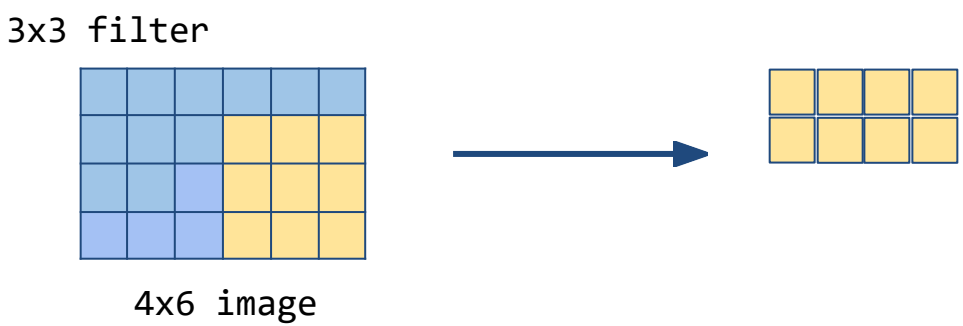

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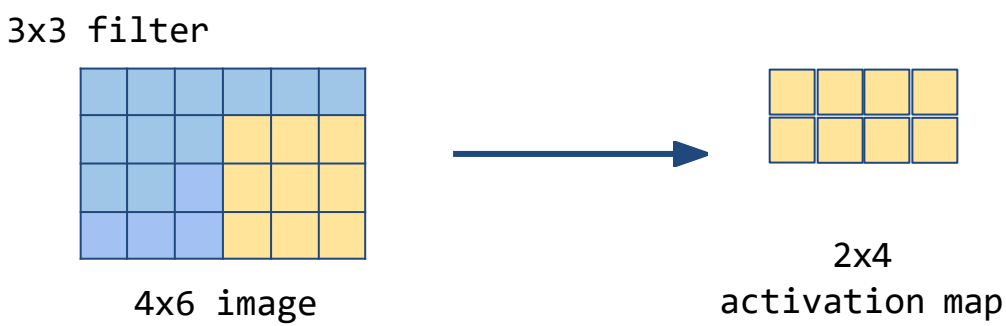

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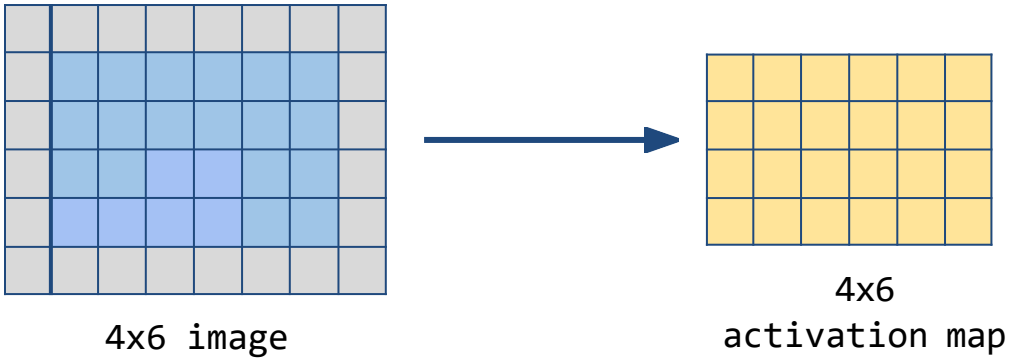
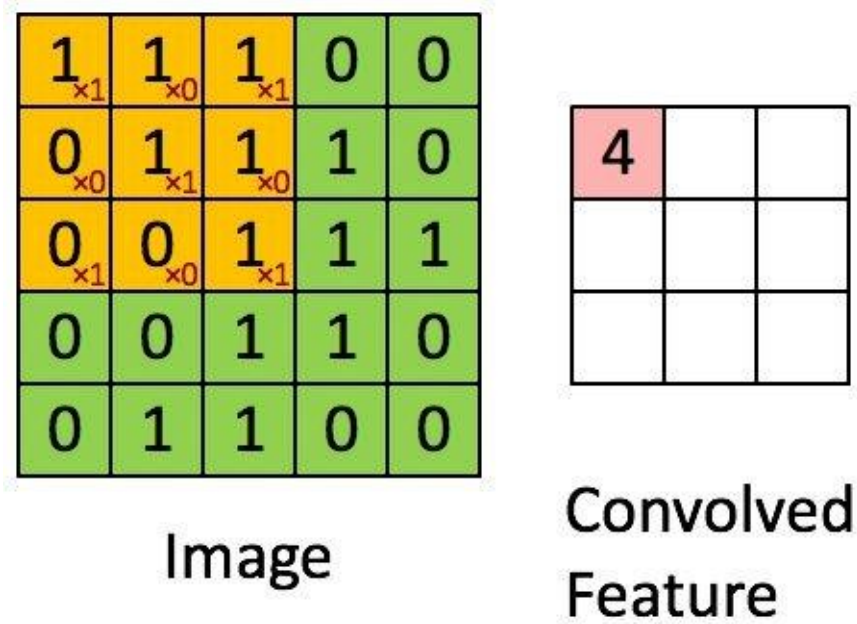
- **Padding:** Each image is padded with “zero” pixels to maintain the image size in the map



Convolutional Neural Networks

- Example of an actual computation



http://deeplearning.stanford.edu/wiki/index.php/Feature_extraction_using_convolution

Convolutional Neural Networks

Convolutional layers in text are quite similar to the ones in images, but we work only in one dimension!

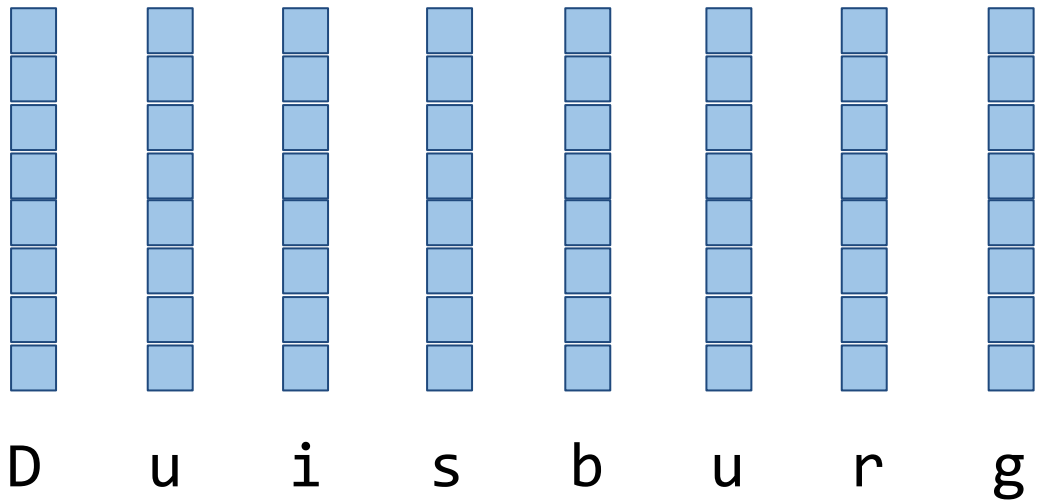
2 Dimensions

John is driving a car

1 Dimension

Convolutional Neural Networks

We “slide” our *filters* on the embeddings of the characters



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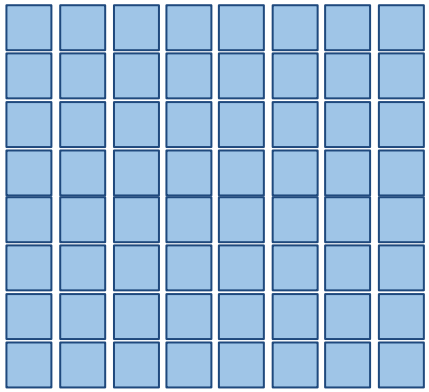

D u i s b u r g

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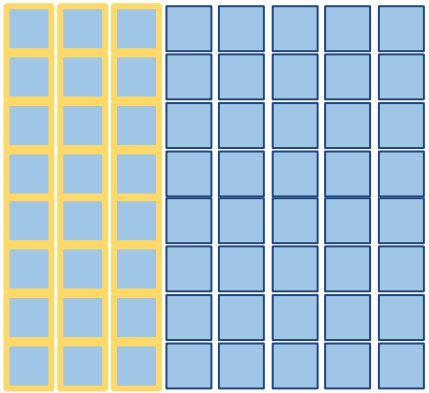

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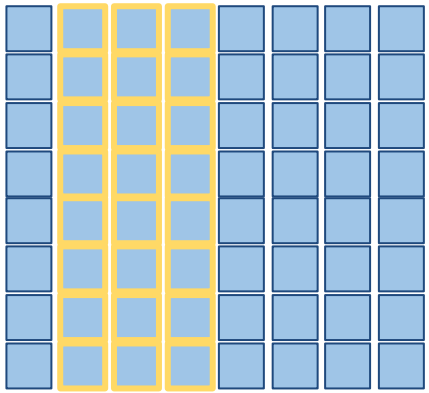

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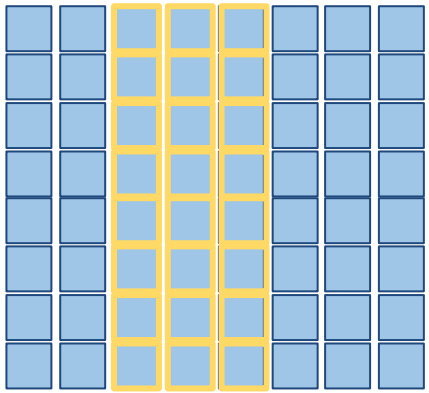

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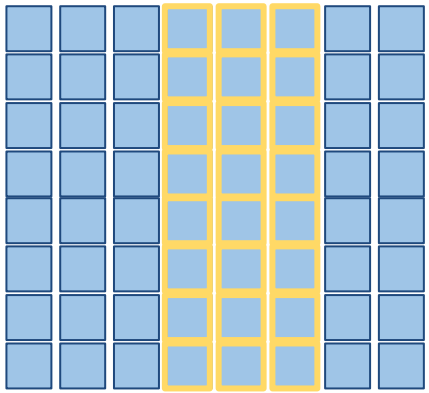

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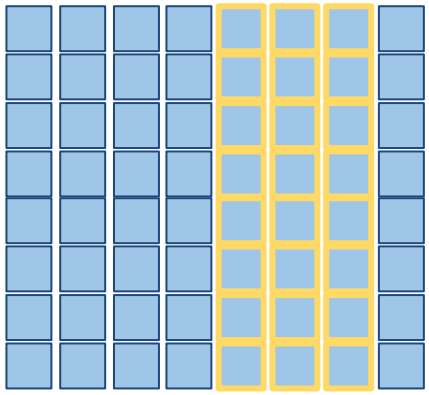

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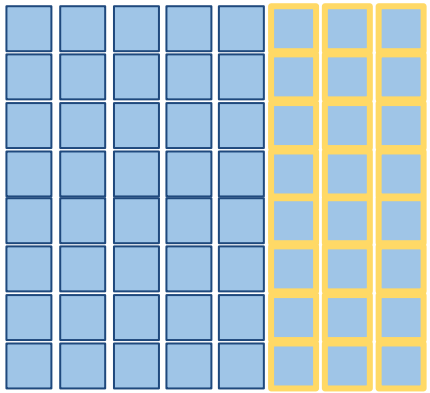
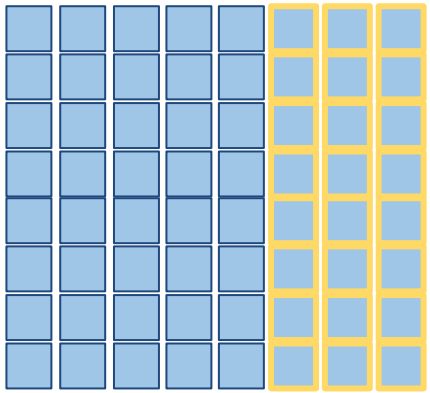


D u i s b u r g

Convolutional Neural Networks

Filters are usually as tall as the embedding layer, and sliding is in one dimension only



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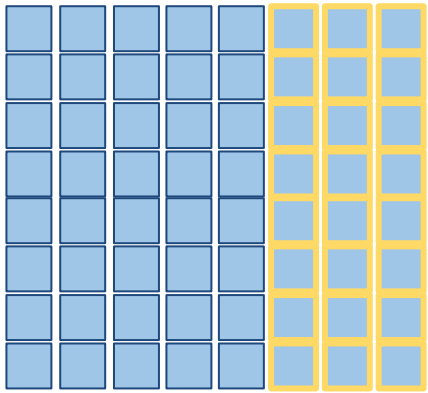
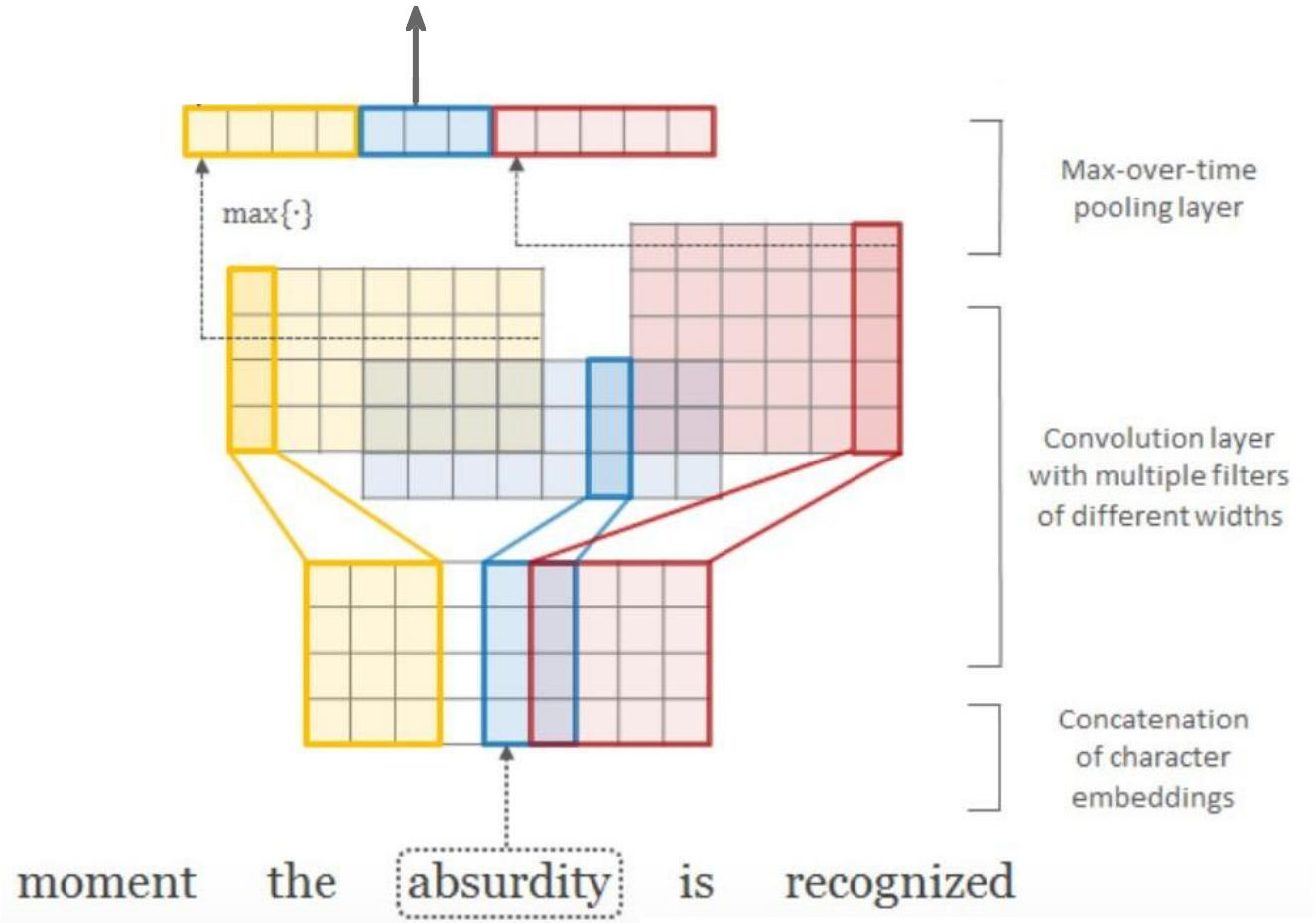


D u i s b u r g

The output in this case will be a 6-vector

Character CNN



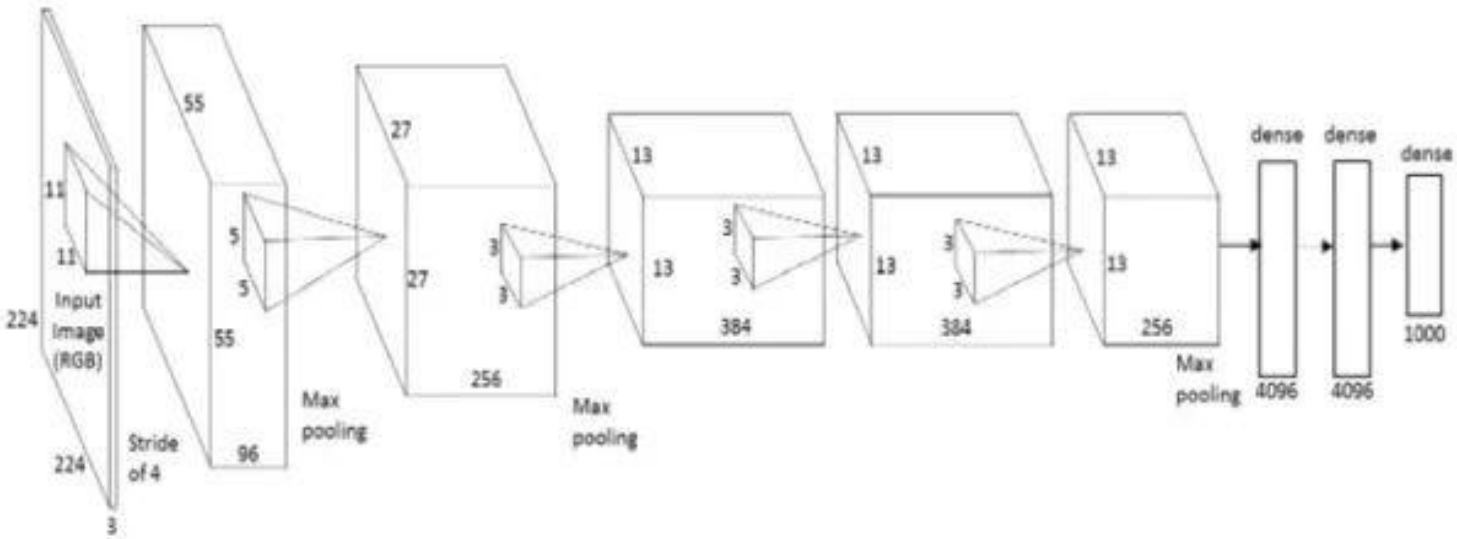
Kim, Jernite, Sontag, Rush, *Character-Aware Neural Language Model*, AAAI 2016

Convolutional Neural Networks

- ConvNets are quite popular in text at character level
- Used to build word embeddings for a larger network

Convolutional Neural Networks

- Atleast in vision, some of the deepest architectures involve ConvNets



Convolutional Neural Networks

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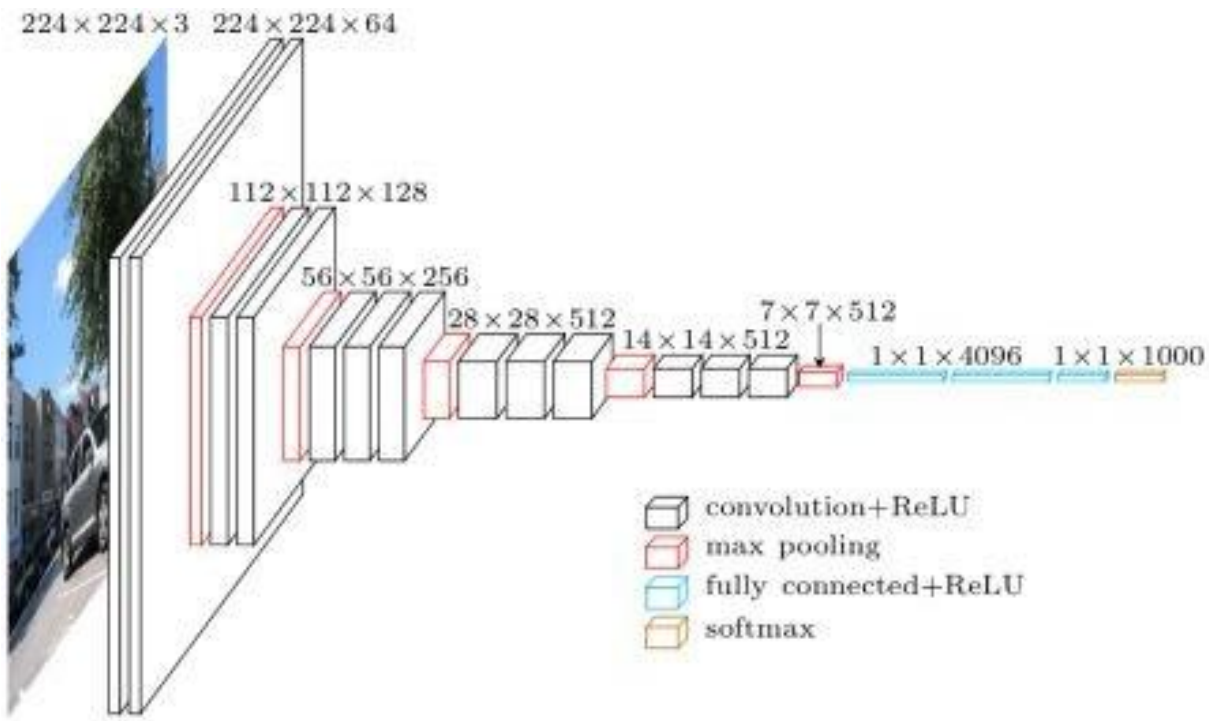

Convolutional Neural Networks

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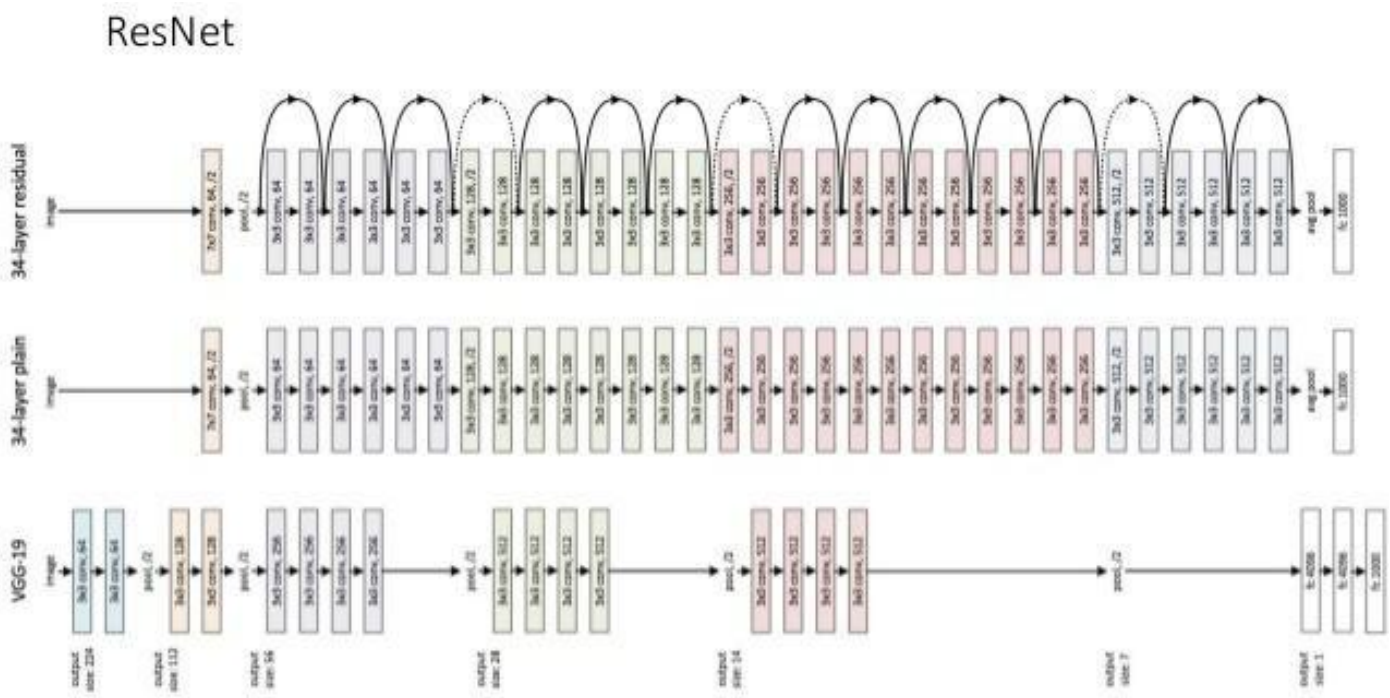
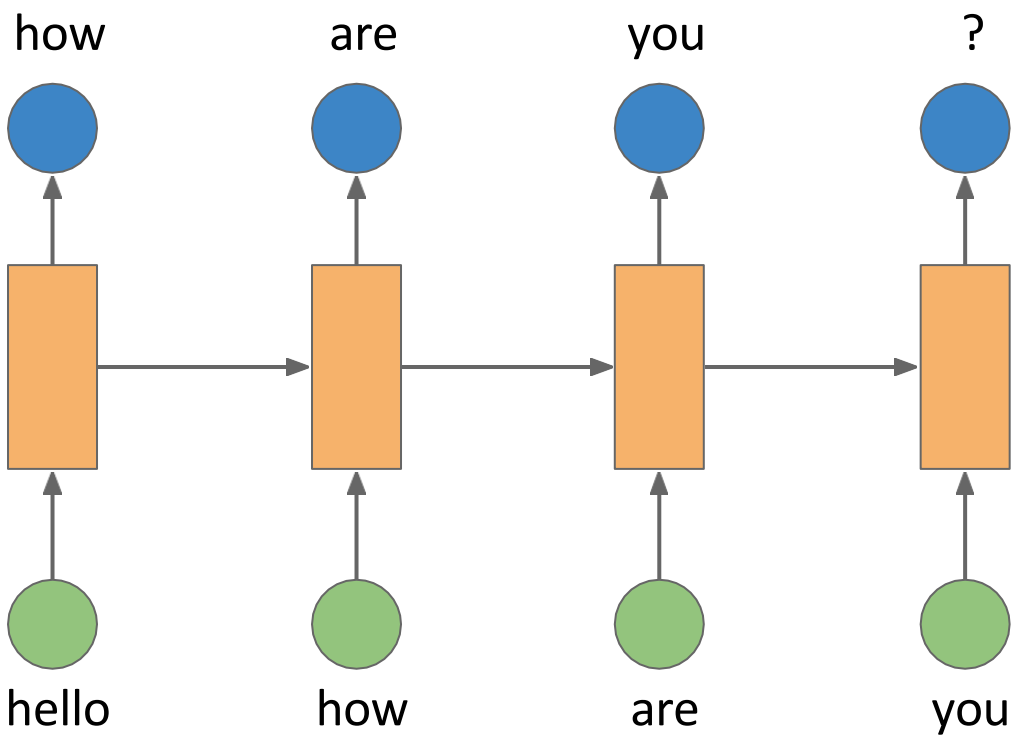


Recursive Neural Network

Recap: Recurrent Neural Network

Summarizes a sentence by reading it word by word in a sequence



Recursive Neural Network

Recurrent neural networks consider a sentence as a flat structure

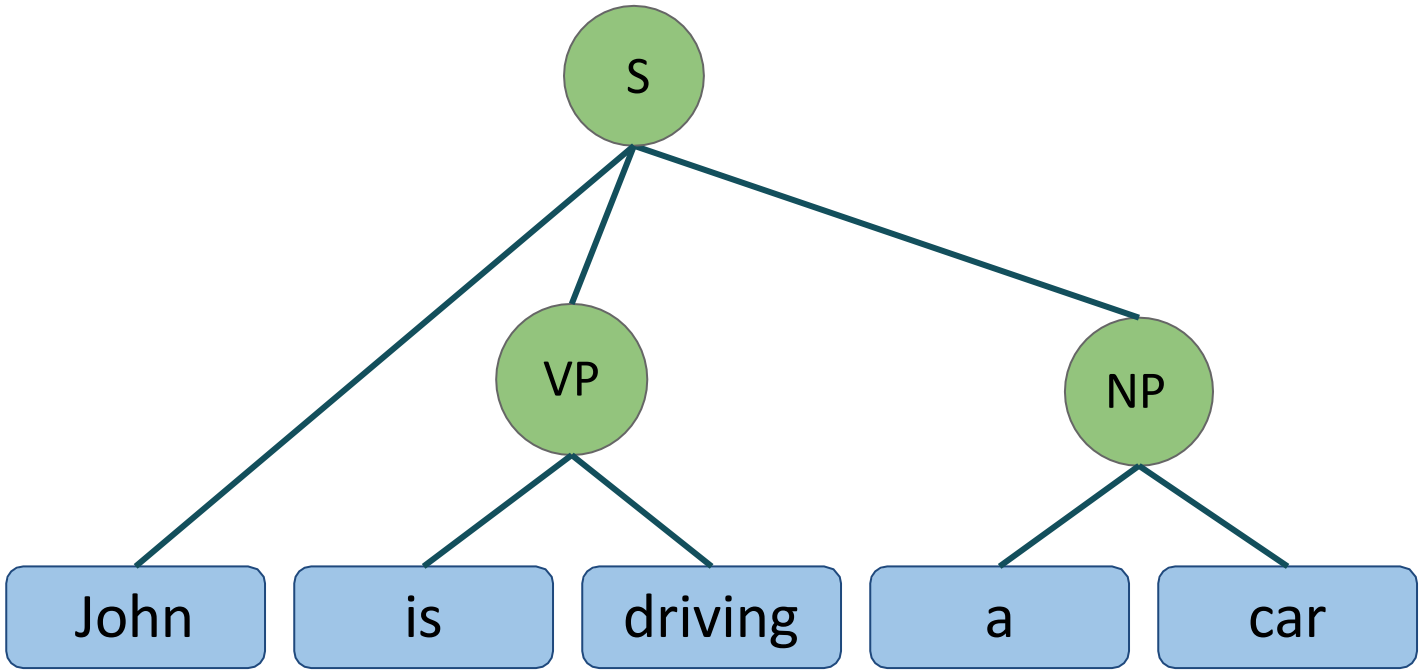
Recursive Neural Network

Recurrent neural networks consider a sentence as a flat structure

However, words in a sentence follow a linguistic hierarchical structure

Recursive Neural Network

Sentences are composed from smaller components - and its easier to see the relationships in this hierarchical structure

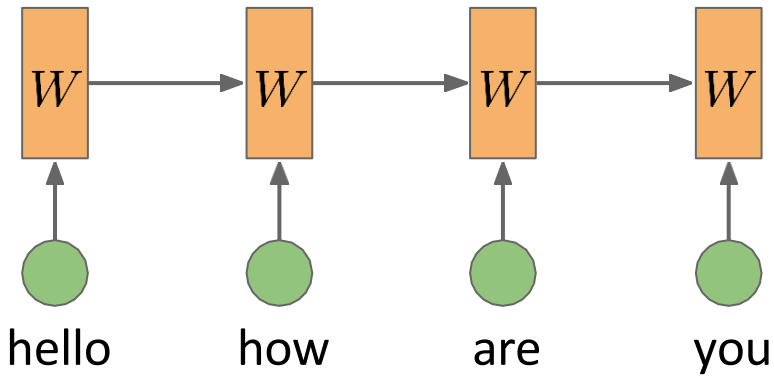


Recursive Neural Network

Recursive NNs are a generalization of recurrent NNs

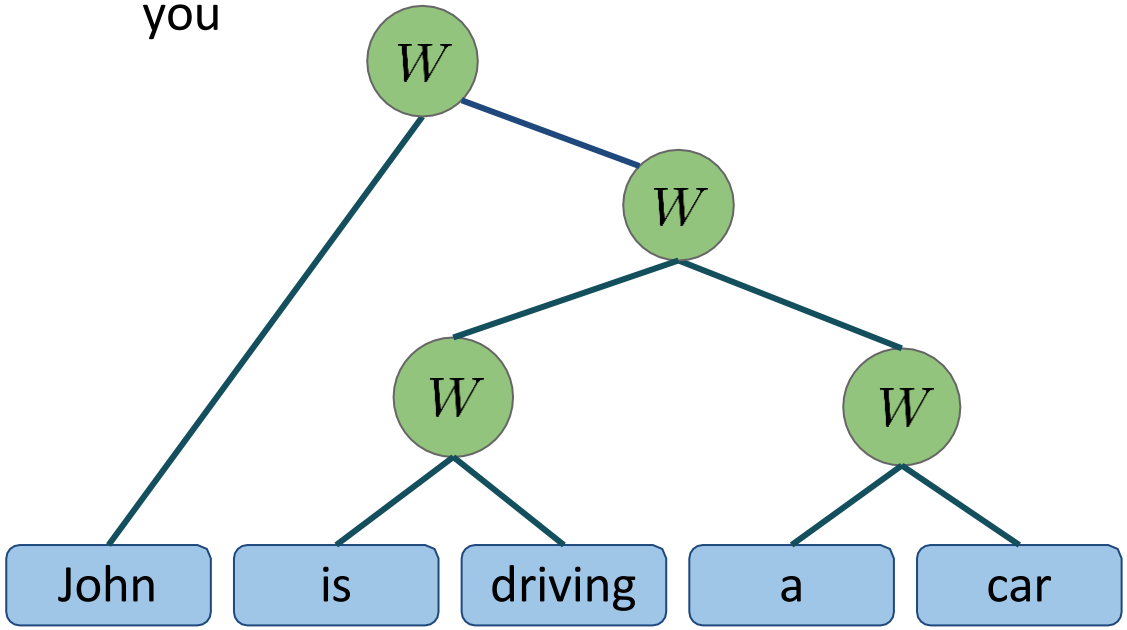
- Instead of reading a sequence from left to right, it reads it inthe form of a tree
- Compared to recurrent NNs, recursive NNs are good in learning long-term dependencies
  - a sequence of length N would result in much smaller sequence (log N) in a hierarchy
- Similar to RNN, identical weight matrix is used for the complete sequence

Recursive Neural Network



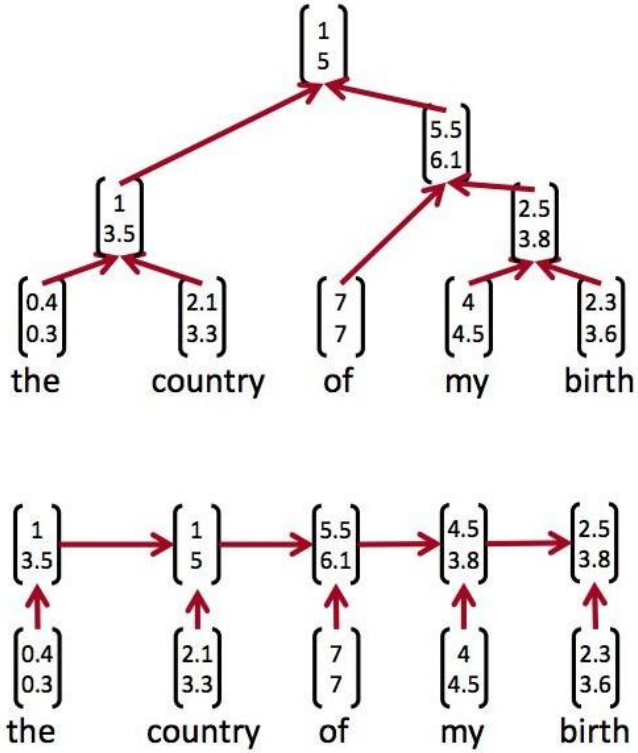
Each recurrent unit takes in one input an the previous hidden state

Each recursive unit can take in multiple inputs



Recursive Neural Network

Recursive vs. recurrent neural networks



Simple Domain Adaptation in Neural Networks

Domain Adaptation

- Training data comes in various styles, genre and domains
  - News, tweets, TED talks, Parliament proceedings
- A system built on one domain may not perform well on other domains
- Several times, annotated data of the domain that we care about (in-domain) is small but there is large heterogeneous out-of-domain data

Domain Adaptation

Domain adaptation aims to use all the available data for the benefit of the in-domain data

Domain Adaptation

Let’s consider a scenario

- When a crisis happens in the world, several NGO’s analyze Twitter feeds to identify needs in the crisis area such as, infrastructure damage, food scarcity
- The NGO’s wait for volunteers to annotate tweets, then build a classifier and automatically classify tweets into classes

Domain Adaptation

- Annotation is a time expensive process in terms of both time and money
- Time is critical especially in the beginning of the crisis

Domain Adaptation

- Annotation is a time expensive process in terms of both time and money
- Time is critical especially in the beginning of the crisis

- Can we make use of the data from previous disaster to speed up the whole process?

Domain Adaptation

Fine tuning / Extended training

- Neural models train in batches
- Once model matures, we can extend training with a different data
- This results in fine-tuning the model parameters towards the new training data
- Essentially, we are resuming the training of an already trained model for a few more epochs but on a different data

Domain Adaptation

- We can train an initial classifier on the previously available data (out-of-domain)
- At the time of a new disaster, this already trained classifier can be used to classify initial tweets
- Once annotated data arrives, we can simply extend the training of our current model on the new data
- Our model parameters will adjust according to the new data

Domain Adaptation

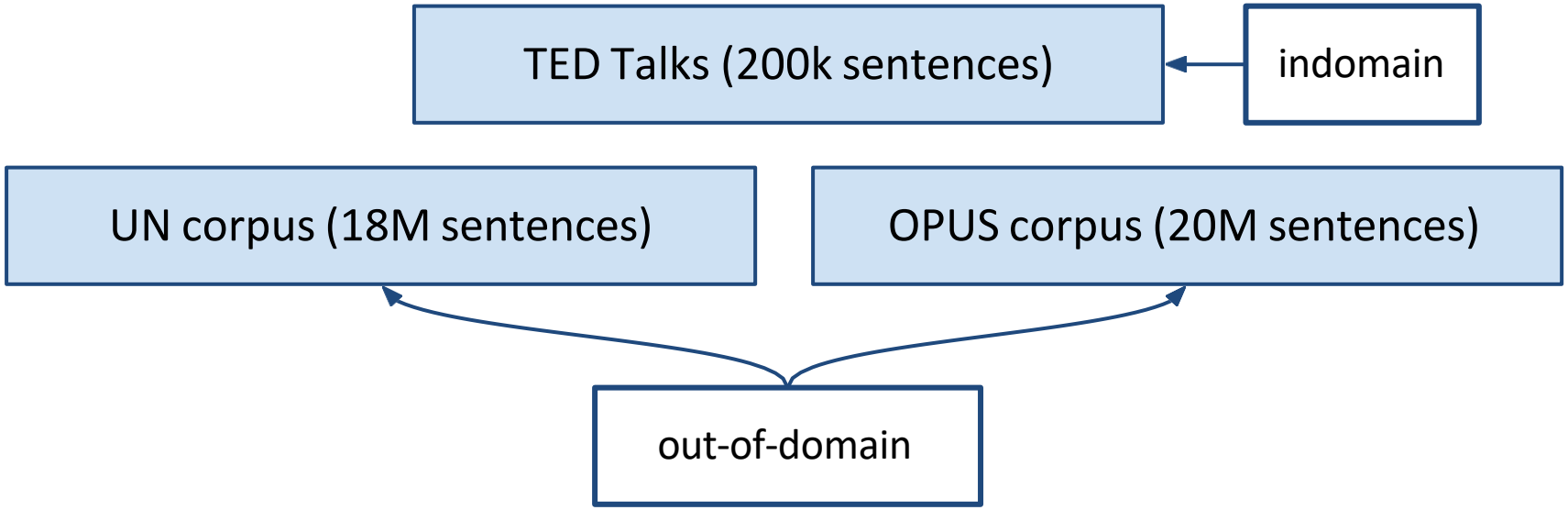
- The performance of the classifier when fine-tuned on the in-domain data improves significantly
- Fine-tuning a model is super efficient compared to training a model from scratch
- The resulting model is robust since it has seen a large amount of out-of-domain data plus some in-domain data.

Domain Adaptation

- Let’s take machine translation as an example

Domain Adaptation

- Suppose, we want to train a neural machine translation (NMT) model for TED talks and there are three corpora available for training



Domain Adaptation

- Suppose, we want to train a neural machine translation (NMT) model for TED talks and there are three corpora available for training

TED Talks (200k sentences)

UN corpus (18M sentences)

OPUS corpus (20M sentences)

- What would be a good training strategy in order to built a good domain specific system?

Domain Adaptation

- What would be a good training strategy in order to built a good TED talks system?
- Use only TED talks because that's what we care about!

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Domain Adaptation

- What would be a good training strategy in order to built a good TED talks system?
- Use only TED talks because that's what we care about!

TED data is small. It would result in overfitting

TED Talks (200k sentences)

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Domain Adaptation

- What would be a good training strategy in order to built a good TED talks system?
- Combine all three corpora together!

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Domain Adaptation

- What would be a good training strategy in order to built a good TED talks system?
- Combine all three corpora together!

TED is comparatively small. It will be lost in the concatenation of three corpora

TED Talks (200k sentences)

UN corpus (18M sentences)

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Domain Adaptation

- What would be a good training strategy in order to built a good TED talks system?

- **Solution:** Train a generic model on all the available data
- Later, **fine-tune** it on the in-domain data

TED Talks (200k sentences)

UN corpus (18M sentences)

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Domain Adaptation - Fine-tuning

- Consider an NMT training scenario that takes an input corpus and trains a model

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Domain Adaptation - Fine-tuning

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- First we train the model on the out-of-domain data for say, 20 epochs
- Now model has learned to translate language from a generic data
- Let’s fine-tune the model to our domain of interest which is TED talks!

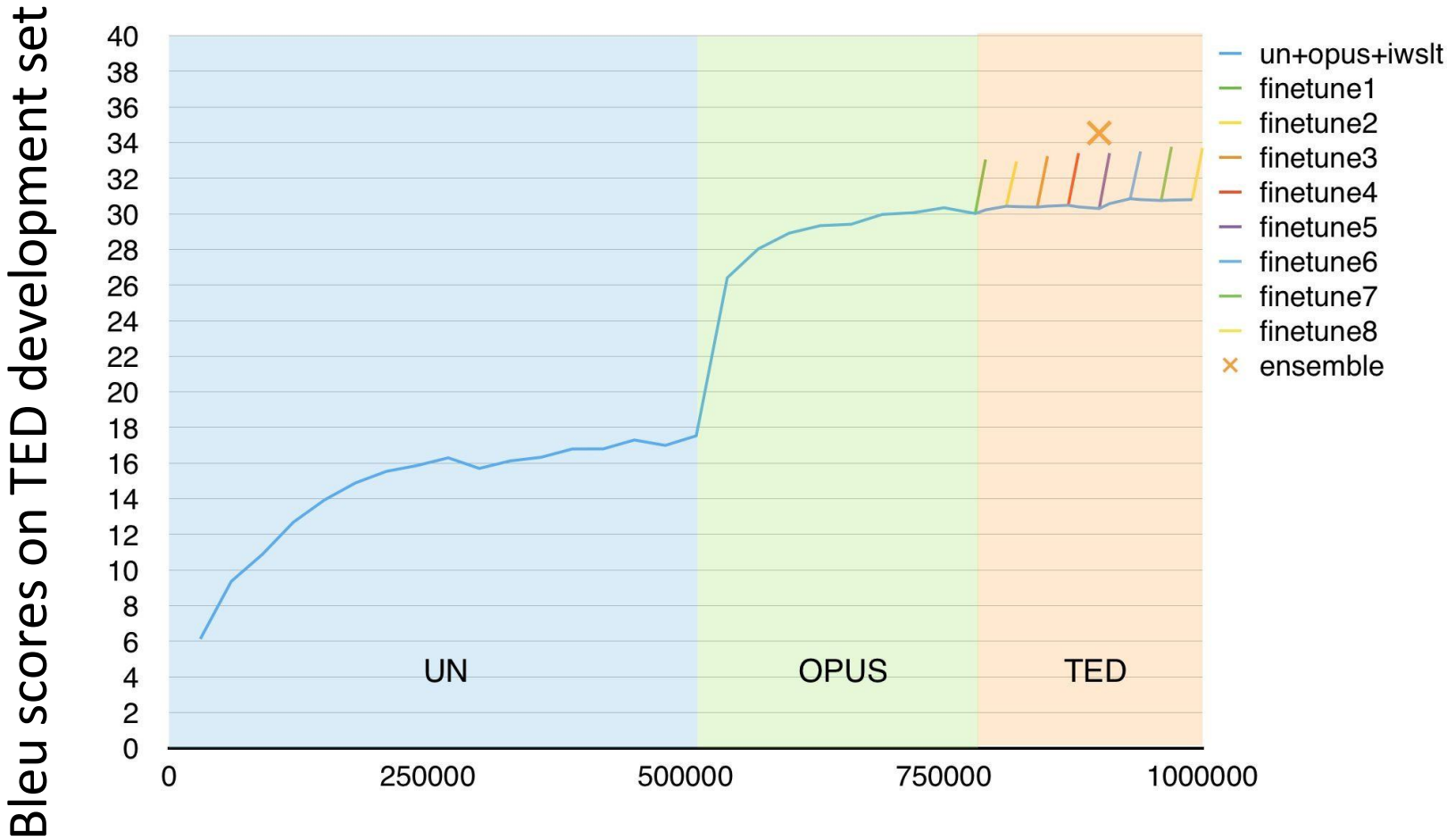
Domain Adaptation - Fine-tuning

- Remember, NMT runs in mini-batches. Essentially we are simply training an NMT system on a few more mini-batches but of TED corpus

Domain Adaptation - Fine-tuning

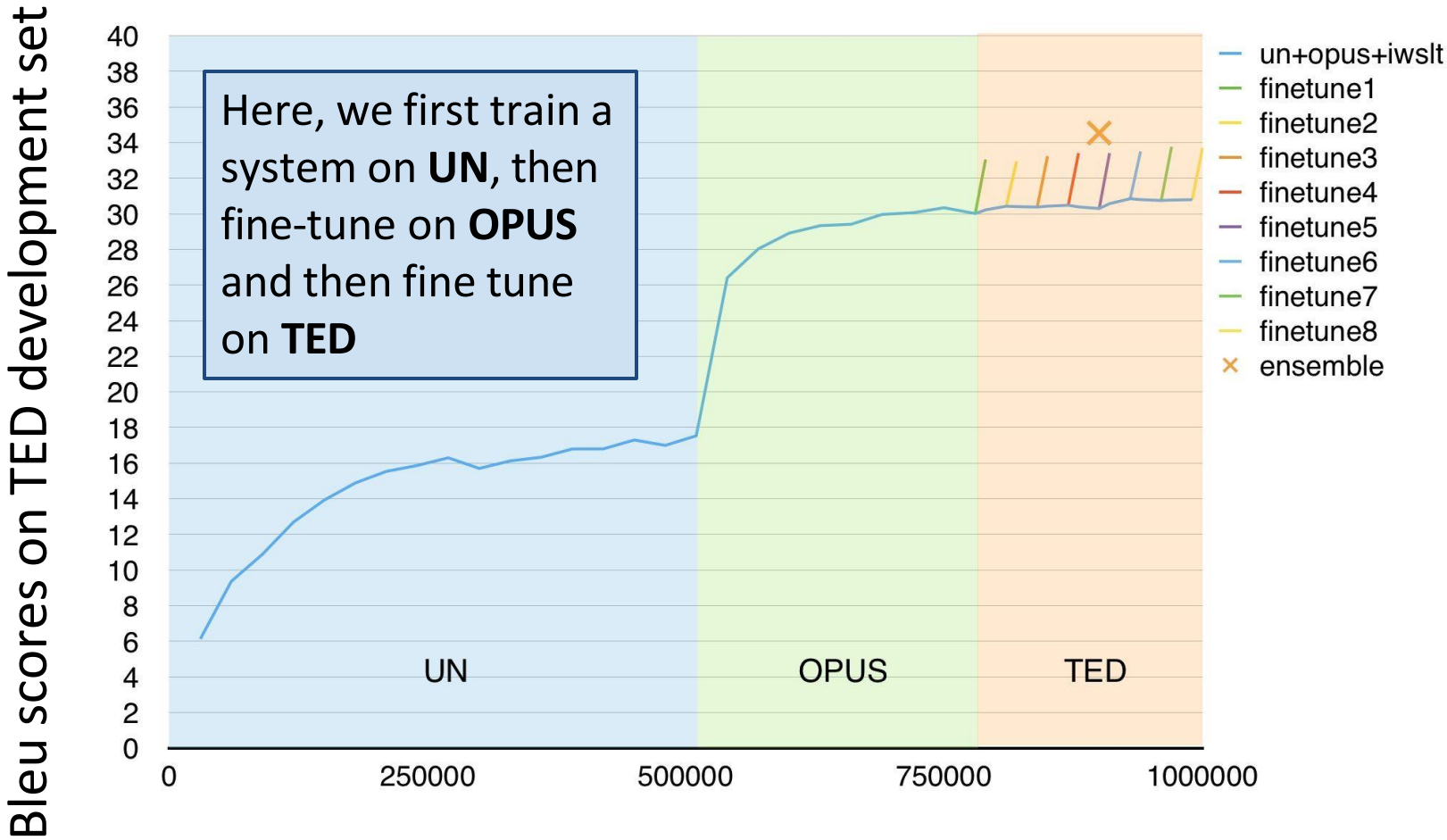
- Remember, NMT runs in mini-batches. Essentially we are simply training an NMT system on a few more mini-batches but of TED corpus
- System learned initial parameters from a large out-of-domain corpora and later we fine-tuned the trained parameters in favor of our in-domain corpus

Domain Adaptation - Fine-tuning

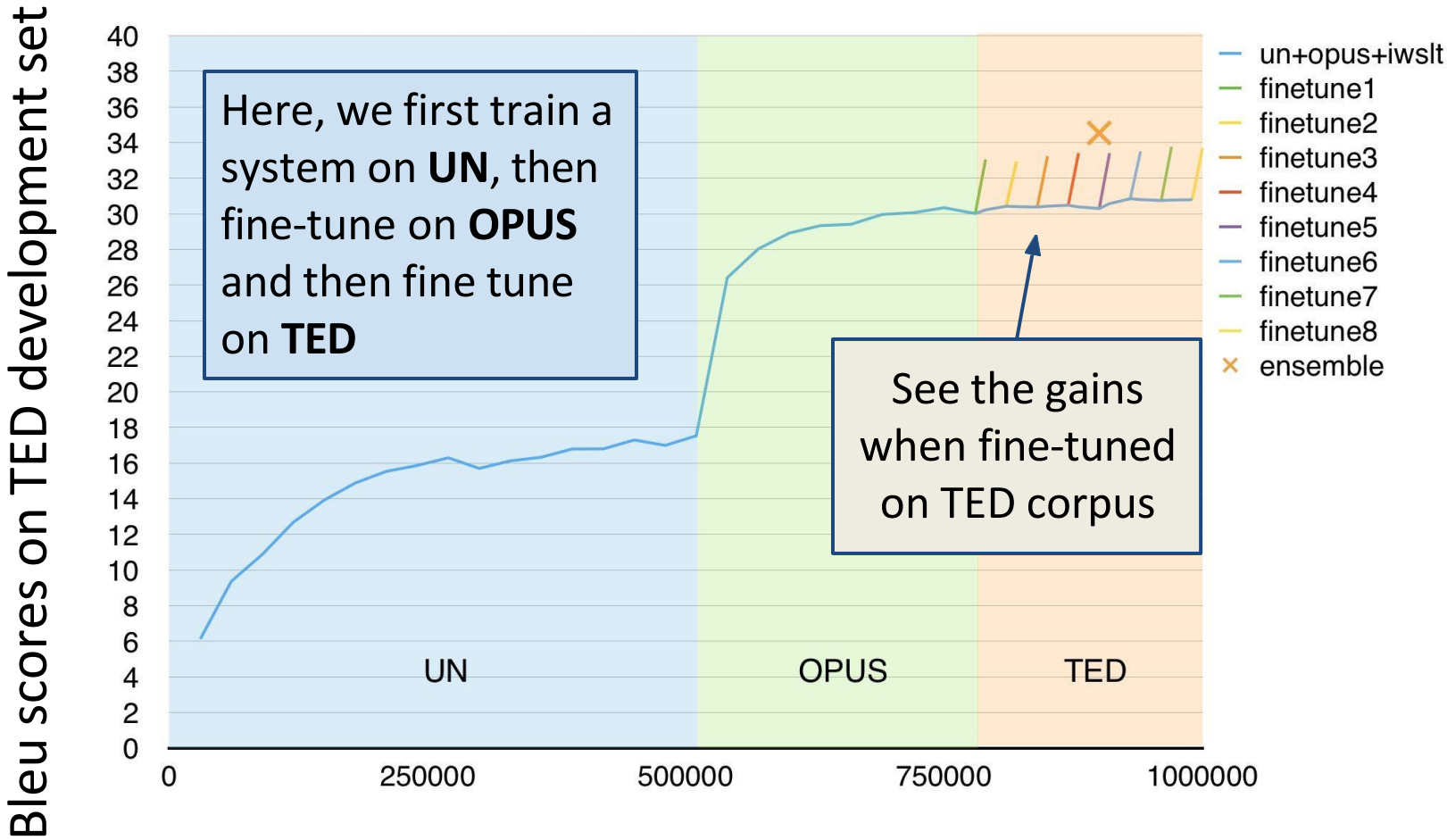


https://arxiv.org/pdf/1708.08712.pdf

Domain Adaptation - Fine-tuning



Domain Adaptation - Fine-tuning



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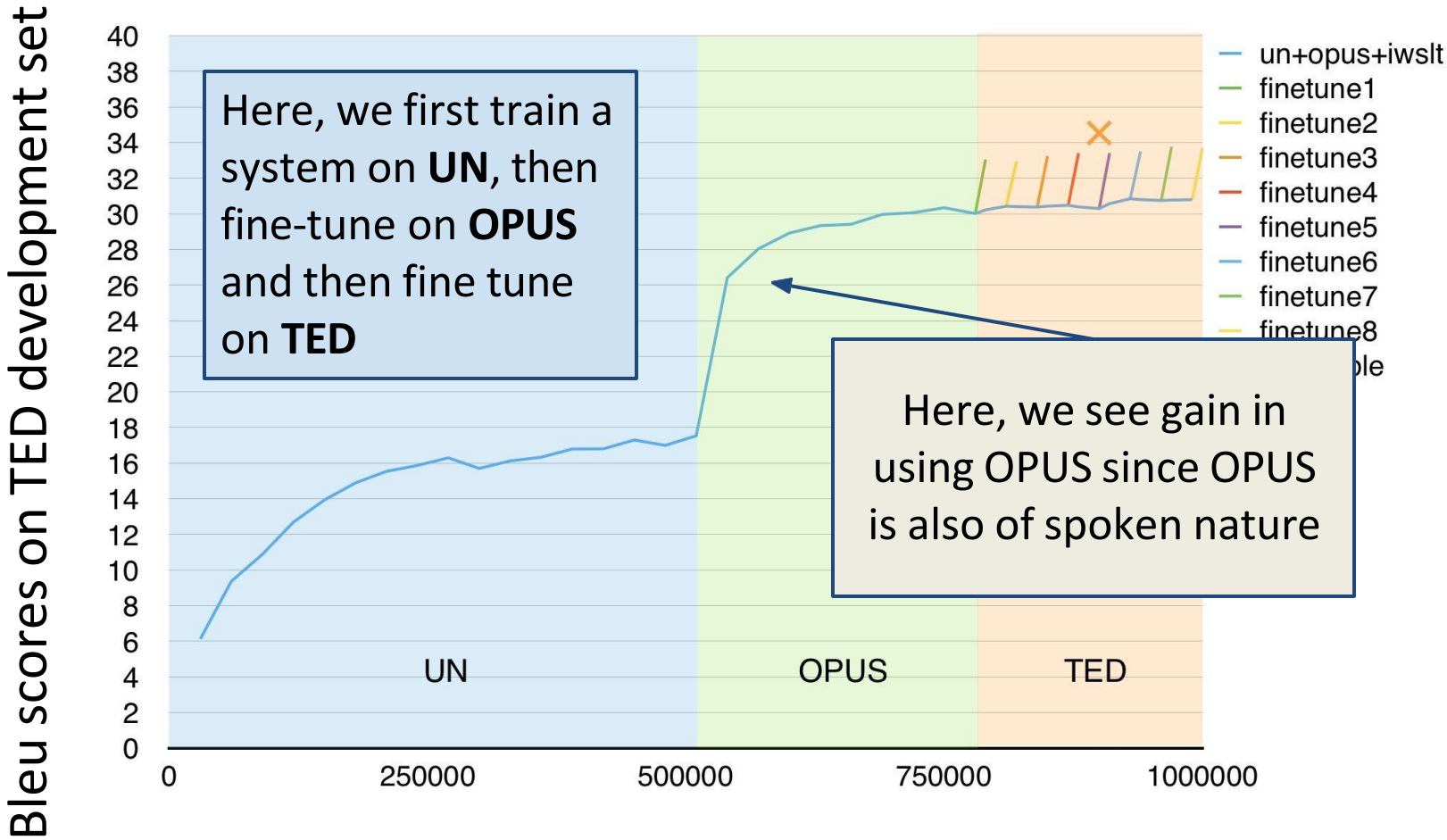


Domain Adaptation

- Fine-tuning is an effective and commonly used method to bias model parameters towards a specific scenario
- Essentially build a general purpose model
- Fine-tune the model towards the domain of interest by resumming the training on the in-domain data

Multi-task Learning

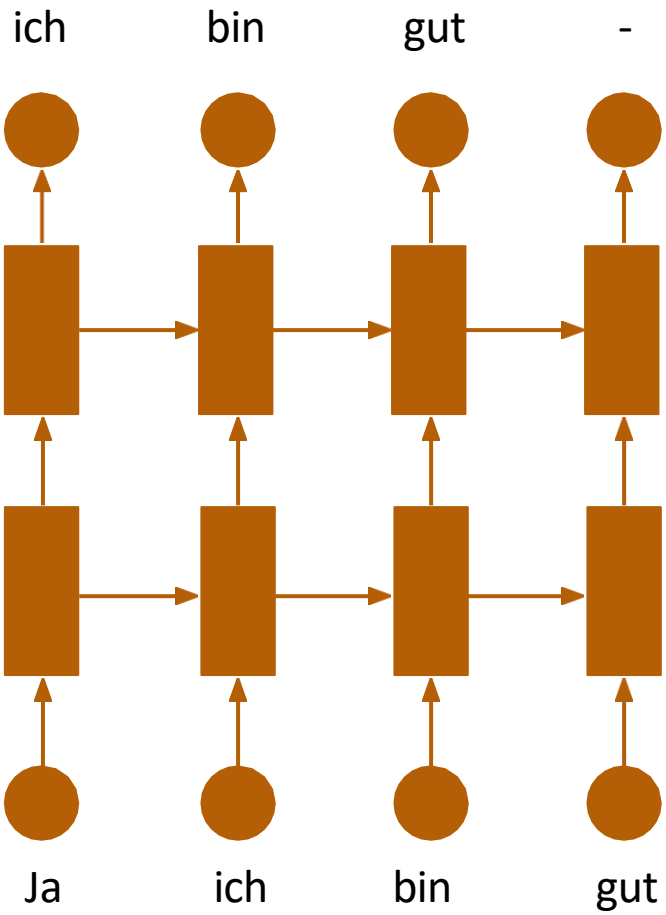
Multi-task Learning

- Learn multiple tasks together
  - Semantic tagging and POS tagging
  - Translation and POS tagging

- Implicitly introducing an inductive bias
  - Model has to consider more than one tasks to reduce loss
- Learning various tasks together helps to achieve better generalization

Multi-task Learning

Language Modeling with part of speech tagging



Multi-task Learning

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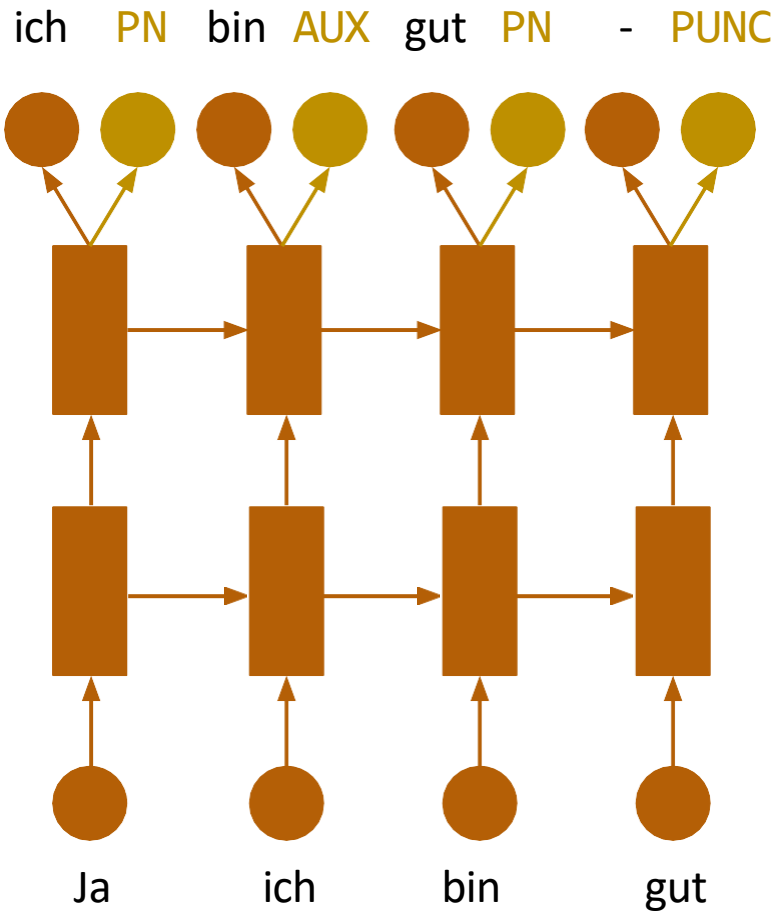


Multi-task Learning

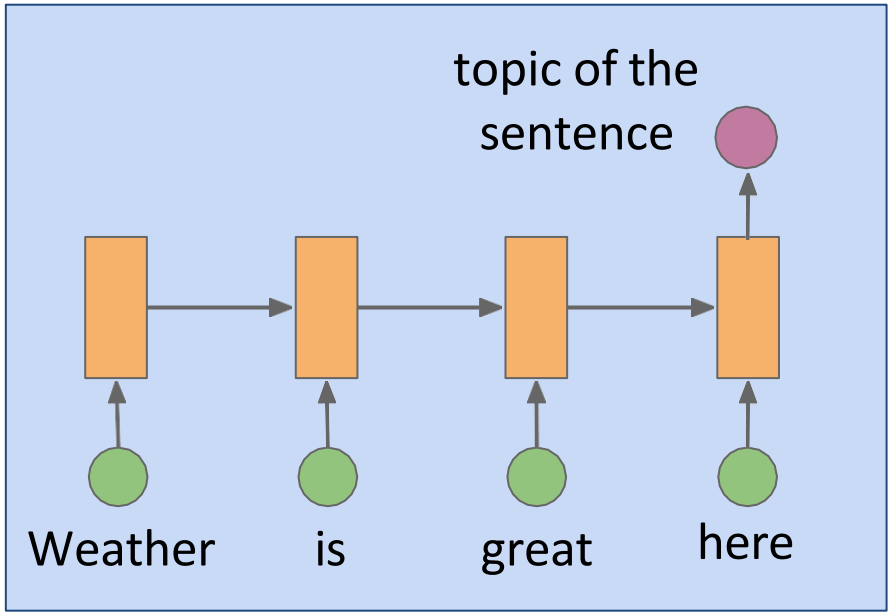
- Input format
  - Data: source sentence
  - Labels: target sentence, POS-tagged sequence
- Since we are learning weights of multiple tasks simultaneously, chances of overfitting are reduced
- As we increase the number of tasks, model has to find weights that minimize the overall loss
- This results in better generalization capability of the model

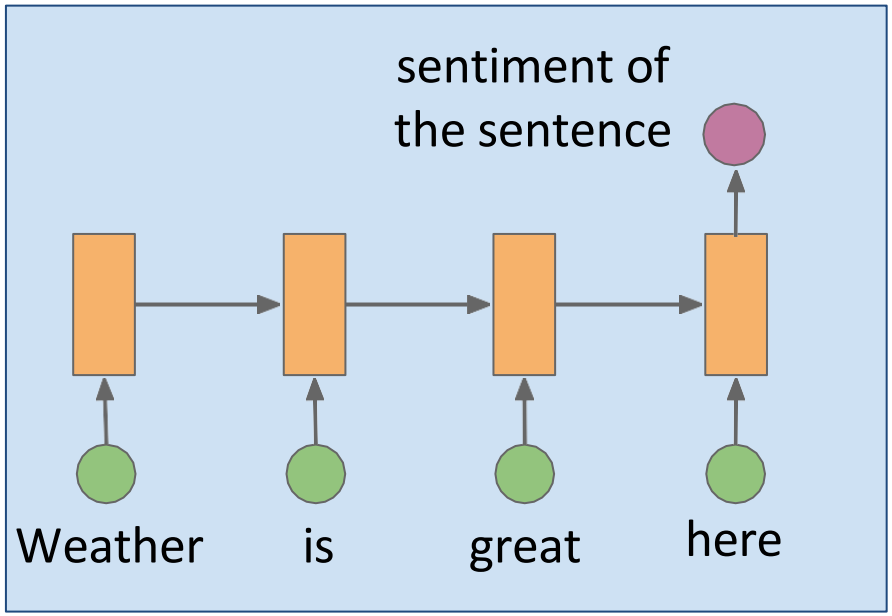
Multi-task Learning

- Let’s take another example to predict topic and sentiment of a sentence using an RNN

Multi-task Learning

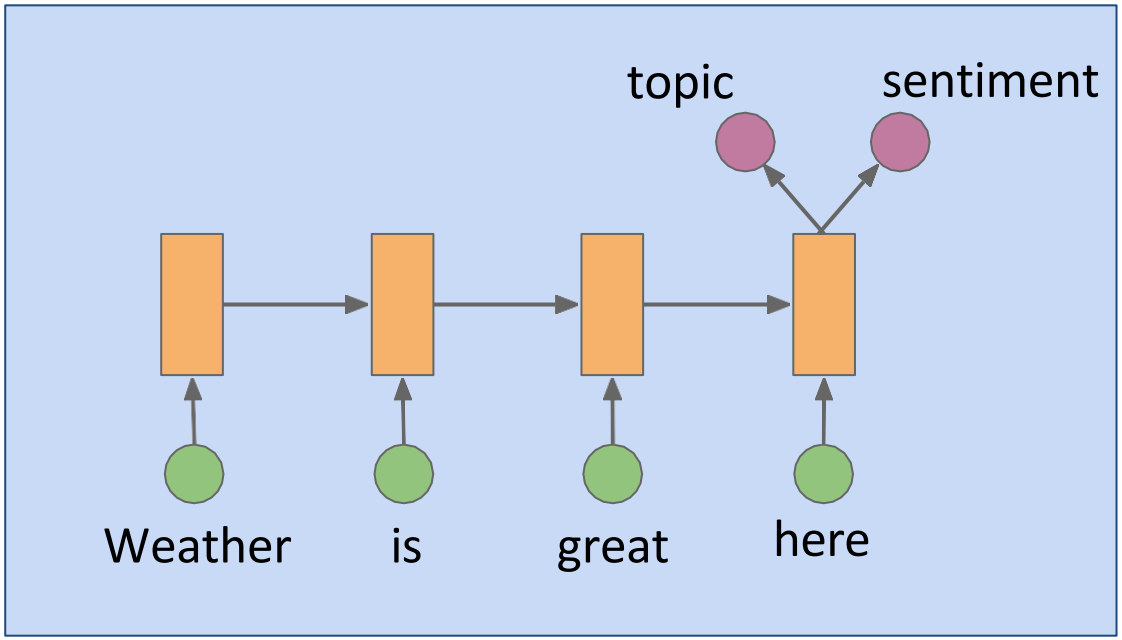
- Let's take another example to predict topic and sentiment of a sentence using an RNN
- We have two models here





Multi-task Learning

- We can combine them into one model that predicts both topic and sentiment of a sentence



Multi-task Learning

- Calculate loss for each task
  - Loss of predicting topic and loss of predicting sentiment
  - Combine them as average loss, weighted average loss, etc.

- In essence, model learns to reduce the loss w.r.t to both tasks

Multi-task Learning

Various architectures to try:

- Shared embeddings
- Network of different sizes
- Various shared and unshared layers

Multi-source Multi-target

Multi-source Multi-target

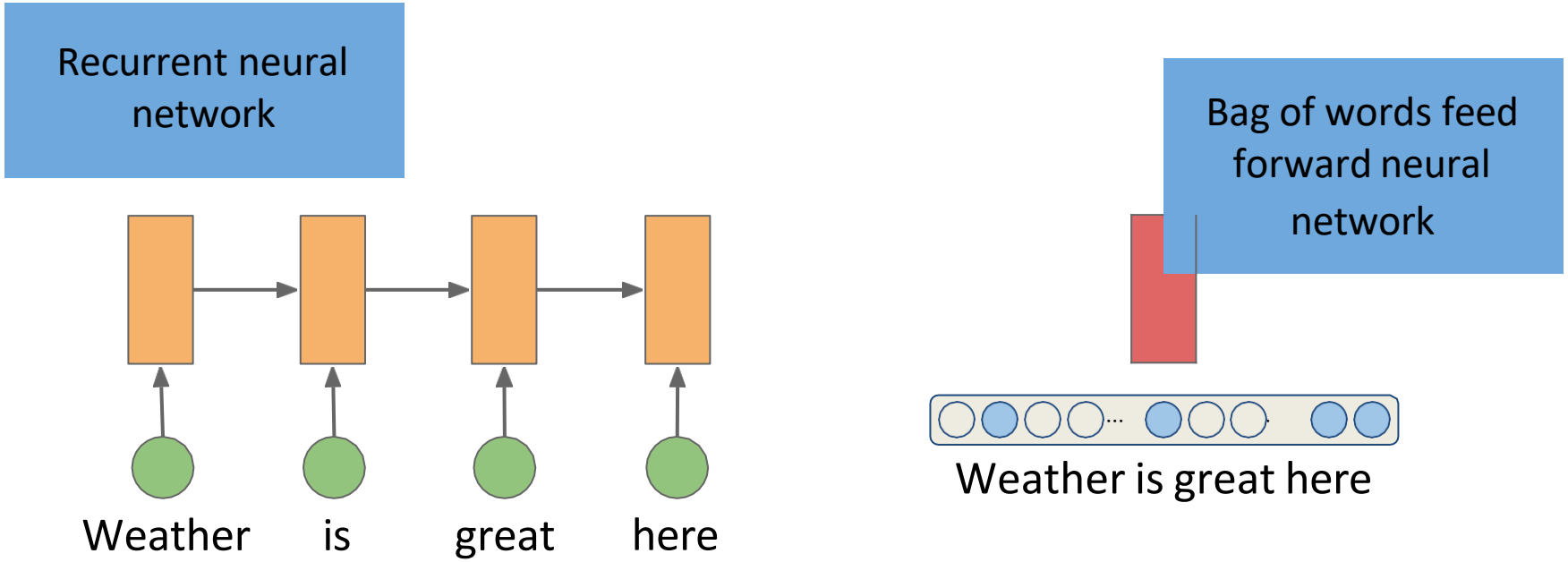
- In multi-task learning, we had one input with multiple output tasks
- One can build a model that takes one or more inputs and learns one or more output tasks
- For example, learn to predict POS tags of related languages together
  - The shared information across languages help to learn a better model
- Another example, learn translation between several closely related languages together

Multi-source Single-target

Let's first look at multi-source single-target model

- Sequence classification task
- We have two networks
  - Feed forward neural network with bag of words as input
  - Recurrent neural network with word sequence as input
- We can combine both networks to perform the classification task

Multi-source Single-target



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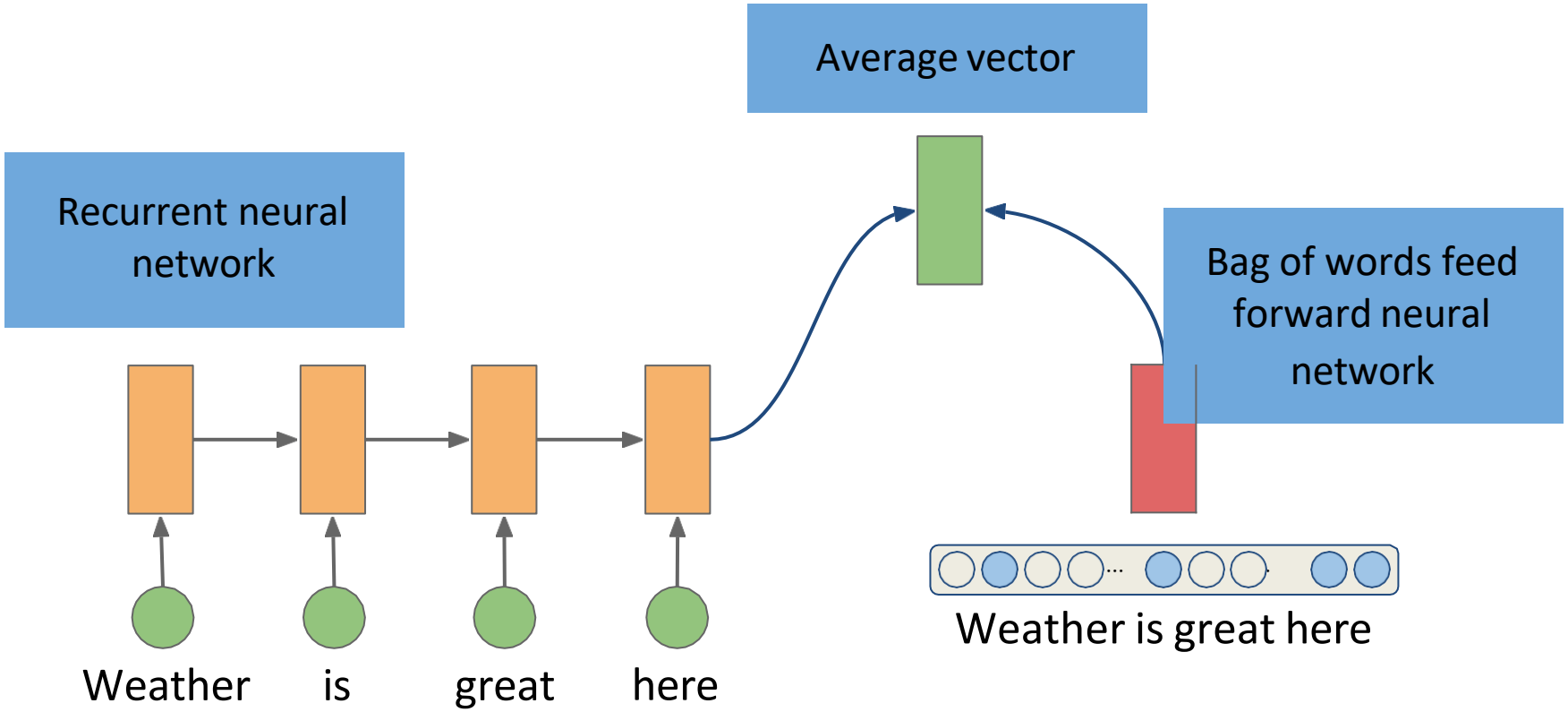

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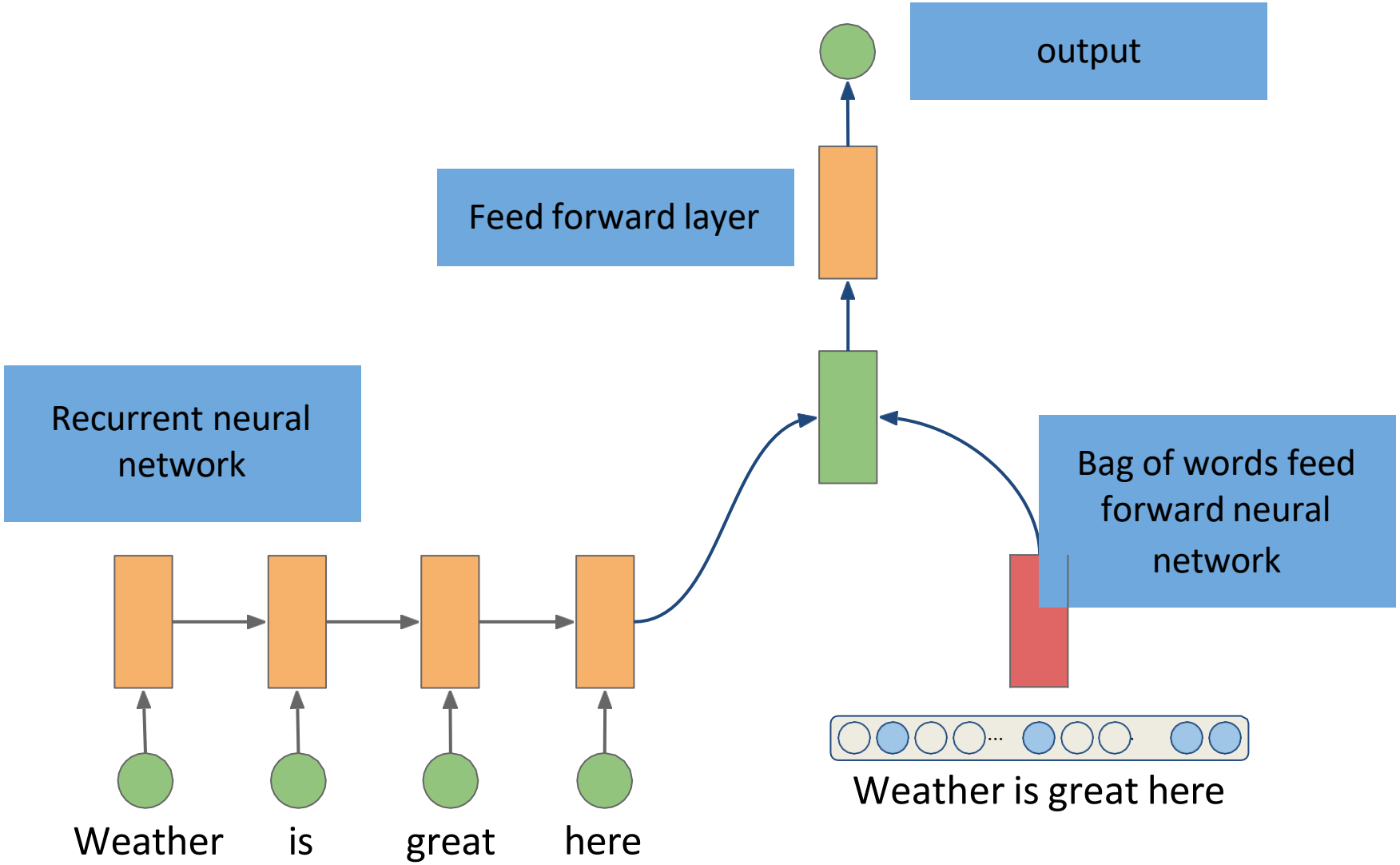


Multi-source Single-target

- The network has a share layer that combines the benefit of both inputs

Multi-source Multi-target

- We can also build networks that take multiple inputs and perform multiple tasks
- Inputs and outputs can be of different types
- These systems benefit from the data of several tasks
- Google NMT is one example of multi-lingual system that is trained on multiple source and target languages

Multi-modal Learning

Multi-modal Learning

Humans tend to learn using various senses e.g. babies use vision and speech together to learn communication

Multi-modal Learning

Humans tend to learn using various senses e.g. babies use vision and speech together to learn communication

We have seen the we can use the same encoder or decoder with more than one language

Multi-modal Learning

Q: As there was no inherent limitation of a single language for the encoder/decoder, is there any inherent limitation for the source/target to be a language at all?

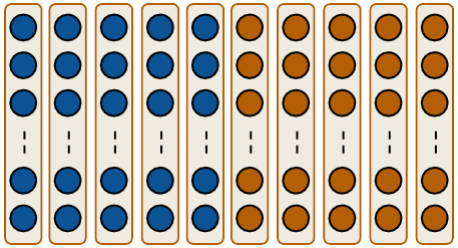
Multi-modal Learning

Q: As there was no inherent limitation of a single language for the encoder/decoder, is there any inherent limitation for the source/target to be a language at all?

A: No! As long as we can teach the machine to convert from any arbitrary input to its intermediate language, we should be able to work with that input!

Multi-modal Learning

Image

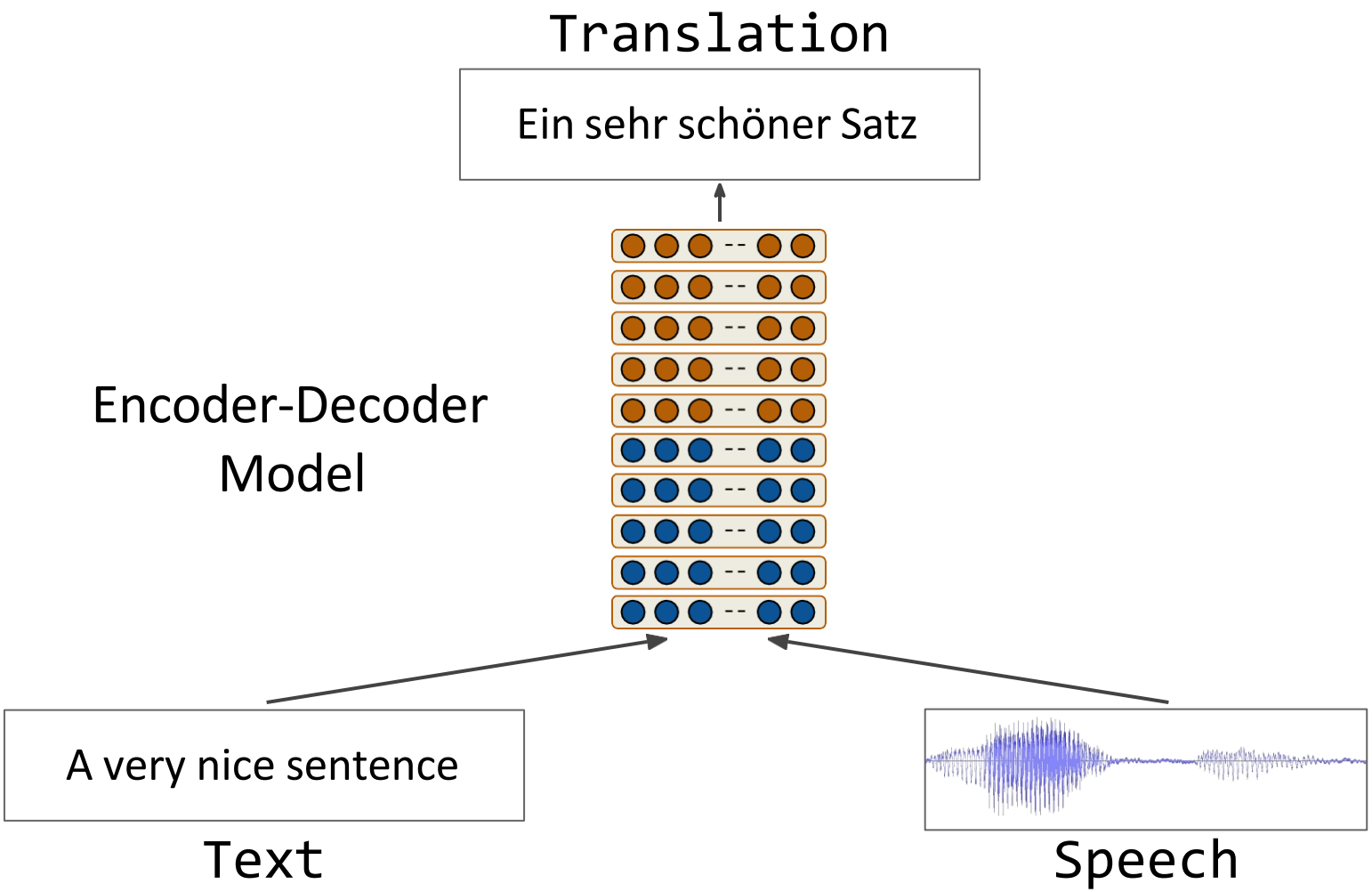


Encoder-Decoder Model

A grumpy cat

Caption

Multi-modal Learning



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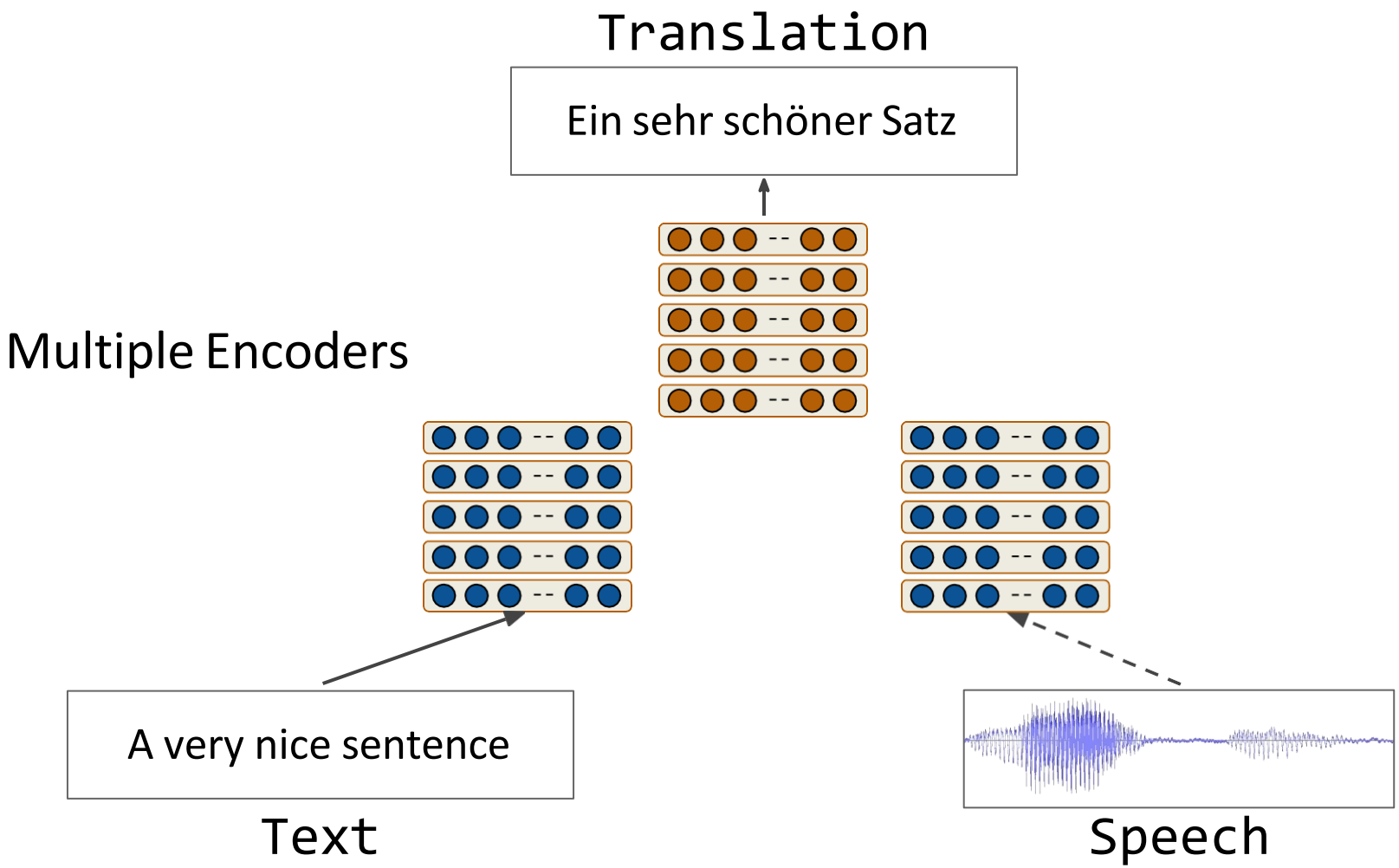

Multimodal Multitask Learning

Each encoder is usually designed for a specific type of input (text, speech, images)

The challenging part is to bring all senses to one space. Jointly training all these tasks with combined losses is one way to do this - still an open problem!

Recent Advancements

Generative Adversarial Networks

Generative Adversarial Networks

So far, we have seen only one class of machine learning problems: Classification problems

Generative Adversarial Networks

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Another class of problems is generating outputs - say generating speech, drawing pictures, producing “handwritten” texts...

Generative Adversarial Networks

We can do this using **Generative Adversarial Networks**!

Humans have evolved over time, and a fundamental motivator for a lot of the progress has been *competition*

Generative Adversarial Networks

We can do this using **Generative Adversarial Networks**!

Humans have evolved over time, and a fundamental motivator for a lot of the progress has been *competition*

Idea: If humans (generally) get better because of *competition*, why not make machines compete with each other as well!

Generative Adversarial Networks

Intuition: We will have two neural networks - one will try and generate something, while the other will try to distinguish if its input is from the real world, or is generated by the first neural network!

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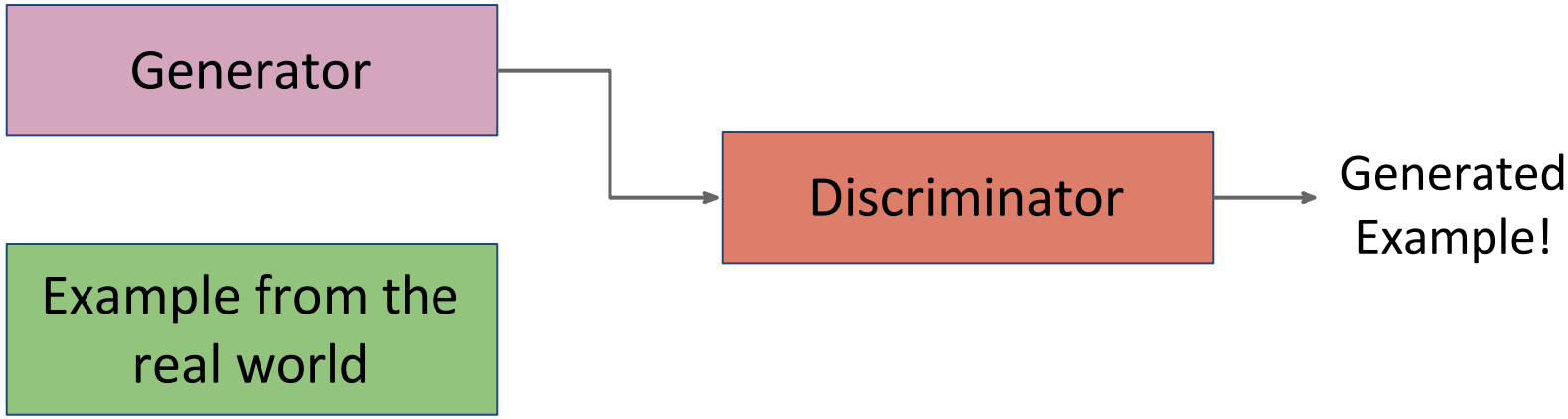

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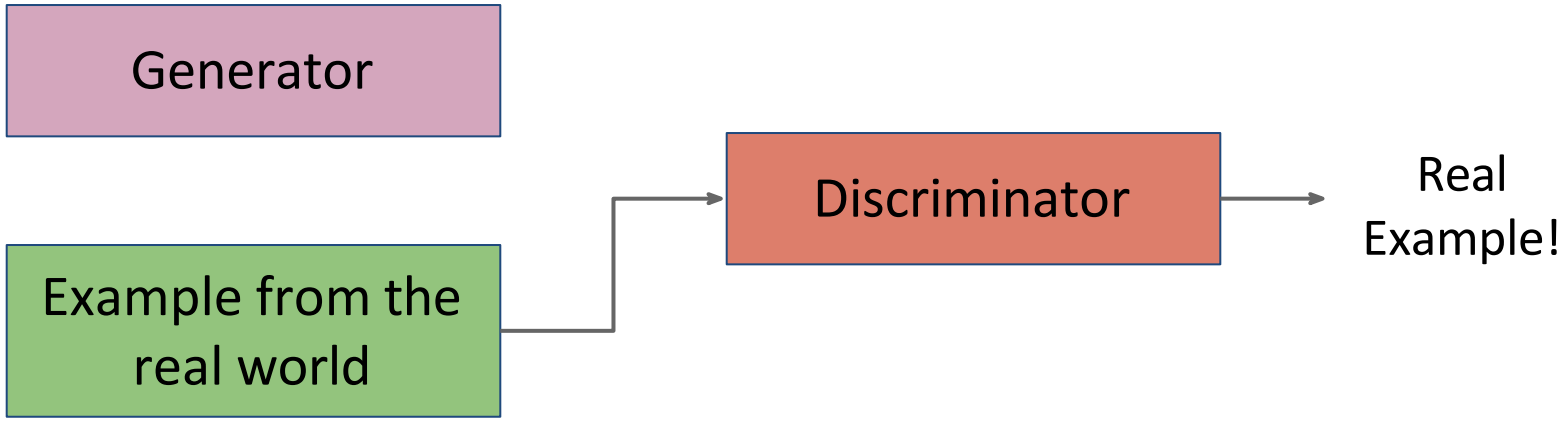
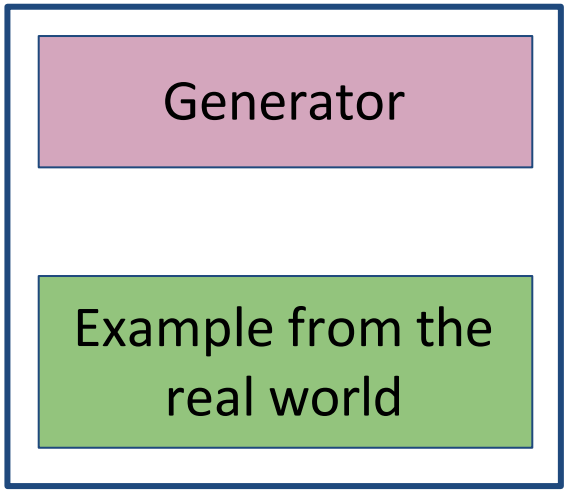


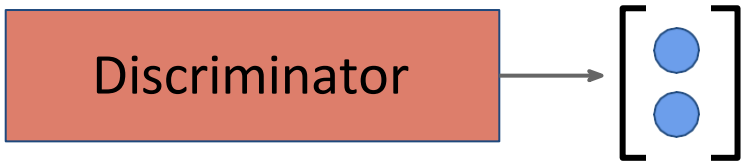
Generative Adversarial Networks

Training process:

1. Freeze Generator and draw outputs from it
2. Draw equal number of outputs from the real world
3. Train the Discriminator on the mix of this data



Freeze and draw outputs



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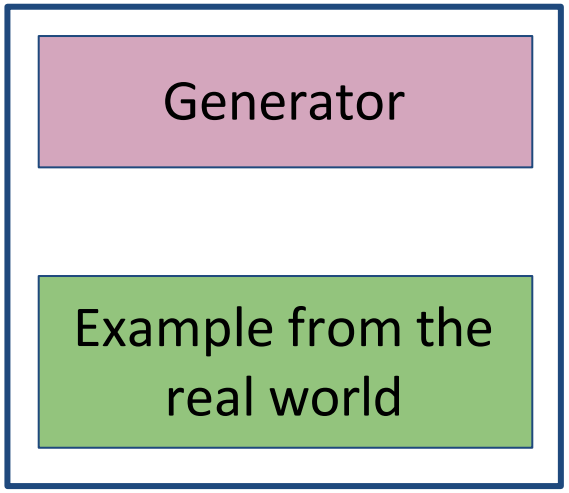
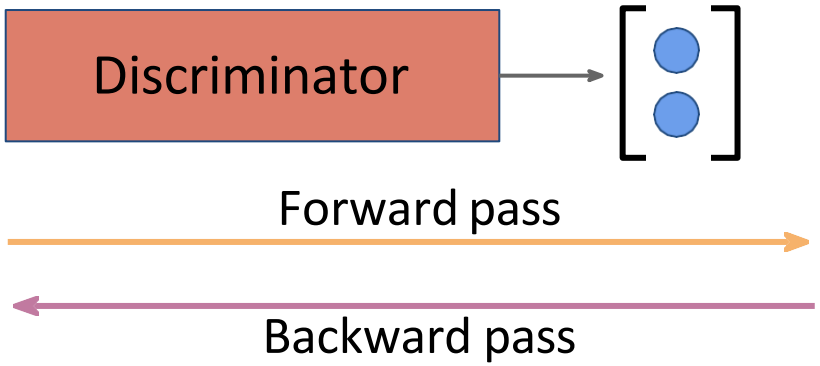


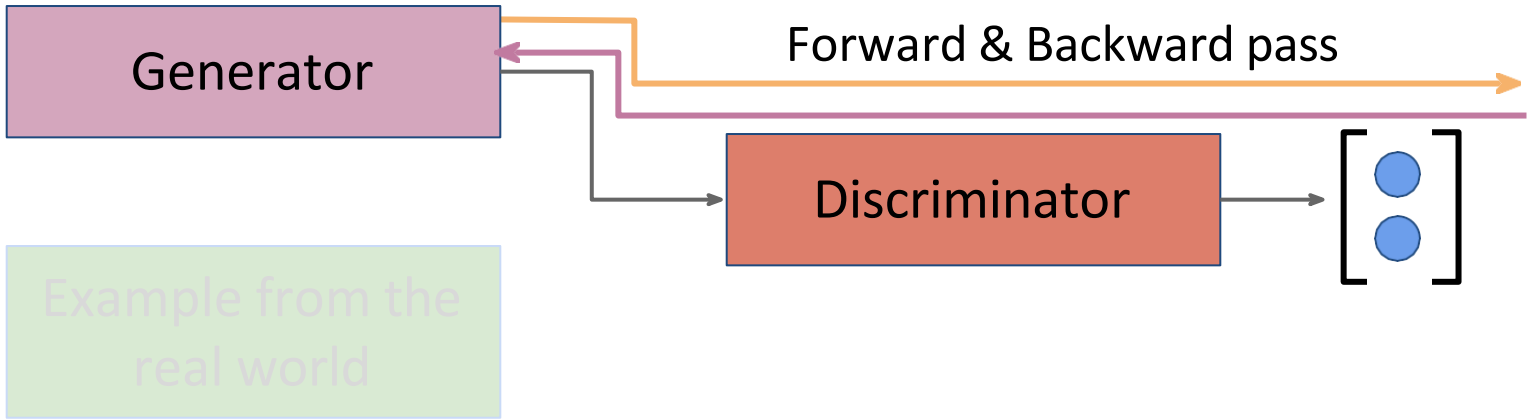
Freeze and draw outputs



Generative Adversarial Networks

Training process:

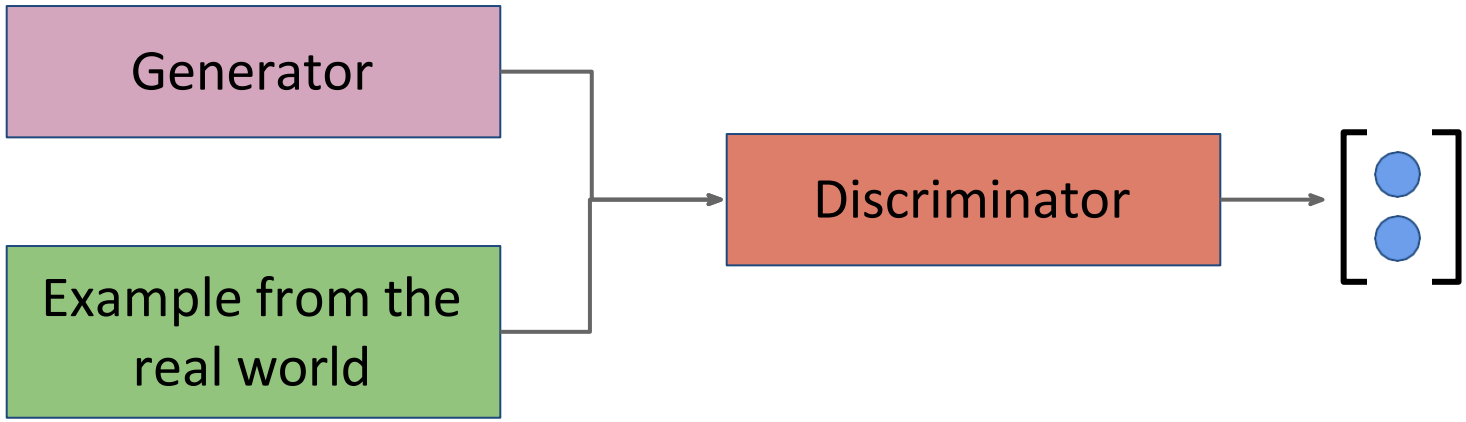
4. Freeze Discriminator
5. Draw an output from the generator
6. Pass the output through the frozen discriminator, and backpropagate the loss!



Generative Adversarial Networks

Training process:

7. Rinse and Repeat steps 1-6
8. Use the generator to generate real-world like examples!



Generative Adversarial Networks

https://www.reddit.com/r/MachineLearning/comments/49wrt4/adversarial_images_for_deep_learning/

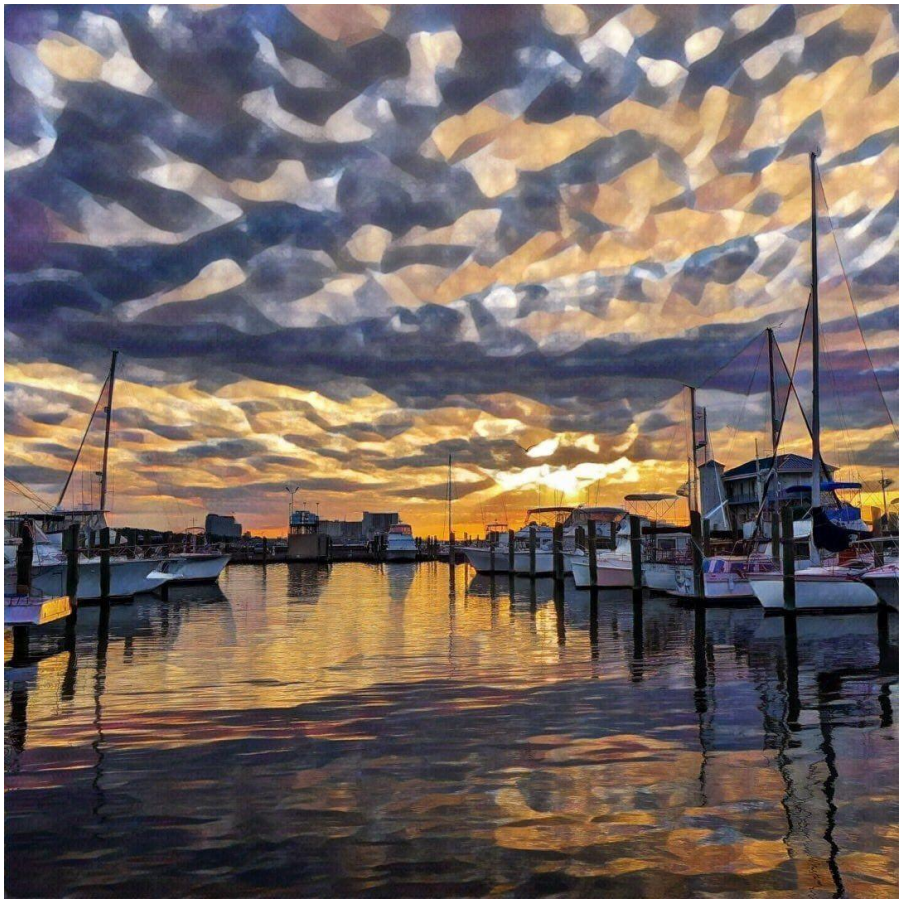
Generative Adversarial Networks

Learning what bedrooms look like:

"Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks"
https://arxiv.org/abs/1511.06434v2

Generative Adversarial Networks

Prisma App: Art from pictures!



https://twitter.com/lapdog99/status/760639333812490240

Generative Adversarial Networks

NIPS 2016 Tutorial: Generative Adversarial Networks
https://arxiv.org/pdf/1701.00160.pdf

Captions to images!

This small blue bird has a short pointy beak and brown on its wings

This bird is completely red with black wings and pointy beak

A small sized bird that has a cream belly and a short pointed bill

A small bird with a black head and wings and features grey wings

Generative Adversarial Networks

NIPS 2016 Tutorial: Generative Adversarial Networks
https://arxiv.org/pdf/1701.00160.pdf

Ofcourse, we also have issues...

Recent Advancements

Reinforcement Learning

Reinforcement Learning

Another class of problems is teaching a machine to perform some **sequence of actions** to **reach an eventual goal**!

Reinforcement Learning

Another class of problems is teaching a machine to perform some **sequence of actions** to **reach an eventual goal**!

Imagine a child's mind when it's learning to walk:

1. The child would notice how adults around it walk
2. It will try to first stand and balance itself - and will fall repeatedly before finding the right "parameters"
3. It will then learn to take small steps. Again, tuning its "parameters" to avoid falling

Reinforcement Learning

This is kind of what reinforcement learning algorithms do!

The task at hand is to train an “agent” who will perform some “actions” and eventually either succeed or fail.

Reinforcement Learning

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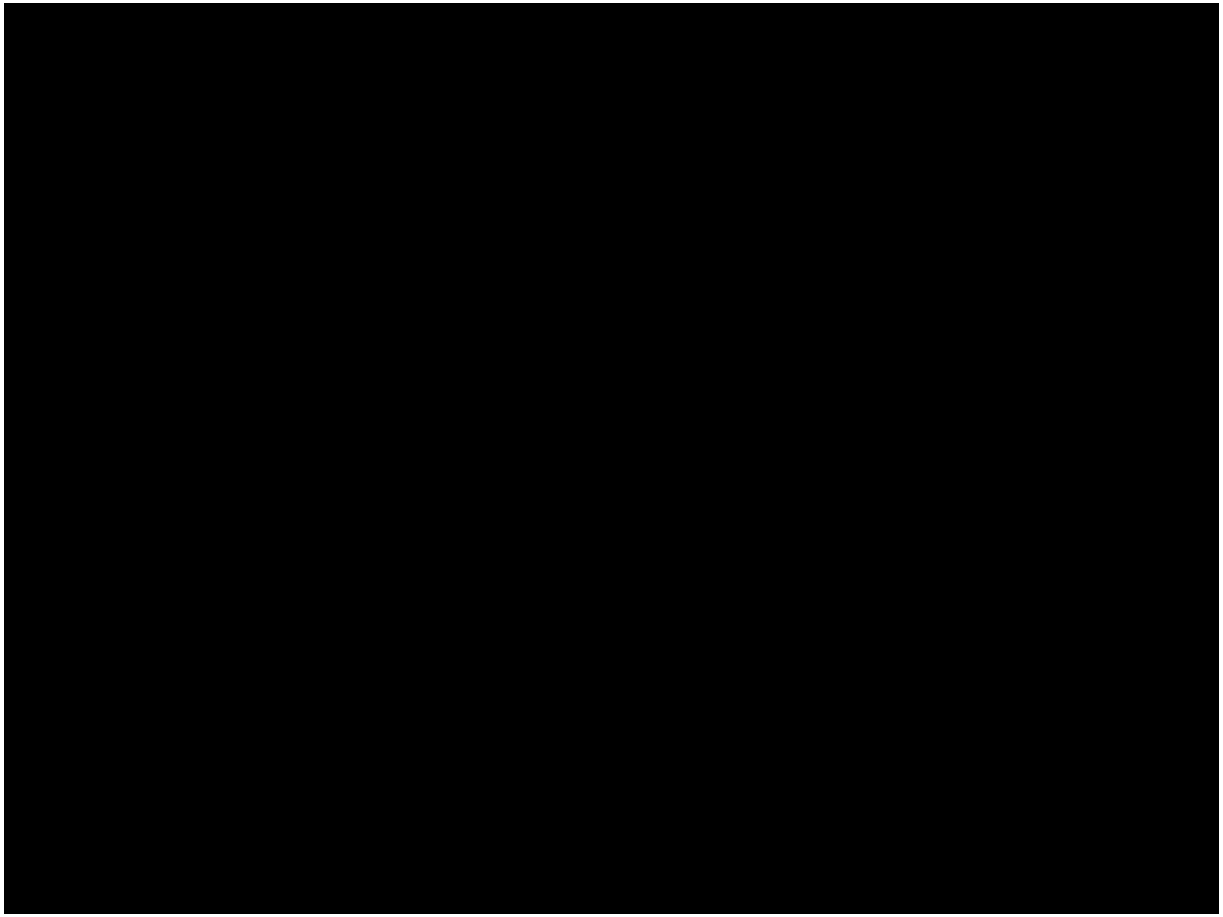
The task at hand is to train an “agent” who will perform some “actions” and eventually either succeed or fail.

Researchers have come up with nice algorithms so that the “agent” can learn from its failures - and eventually succeed

Reinforcement Learning

But first, an example!



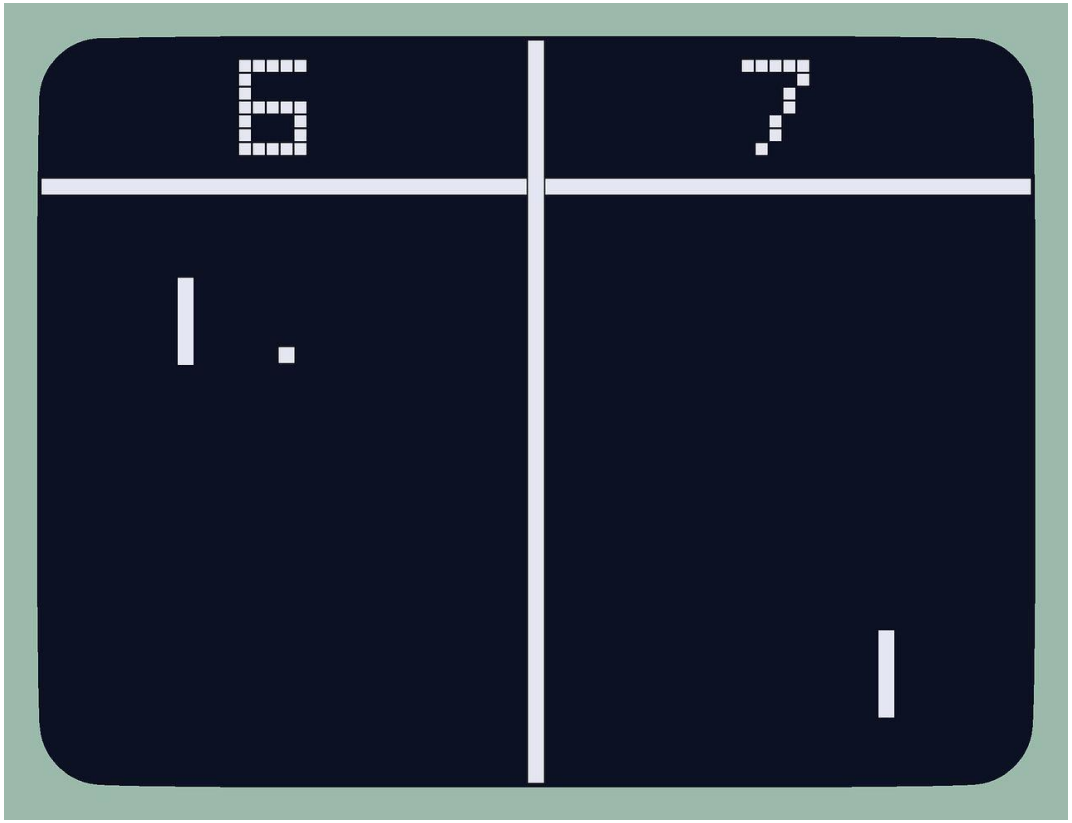
http://heli.stanford.edu/

Reinforcement Learning

A simplified walkthrough: Playing Pong!

Reinforcement Learning

Goal: We want to teach our computer to play the game of Pong



Reinforcement Learning

Goal: We want to teach our computer to play the game of Pong

General Idea: At each timestep (say every 100ms), we want to see where the ball is, and decide if we want to move our paddle up or down

Reinforcement Learning

Training process:

1. Initially, we will move our paddle up or down randomly, and most probably fail

Reinforcement Learning

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2. Before failing, we have taken a set of actions

```
[up, down, up, up, up, down, down, up, down …]
```

Reinforcement Learning

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`[up, down, up, up, up, down, down, up, down ...]`

3. Some action(s) caused us to eventually fail, so let us penalize all these actions a little bit (In some variations, penalize recent actions more)

Reinforcement Learning

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4. Every time we succeed, boost all the actions a little bit

Reinforcement Learning

Training process:

1. Initially, we will move our paddle up or down randomly, and most probably fail

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`[up, down, up, up, up, down, down, up, down ...]`

3. Some action(s) caused us to eventually fail, so let us penalize all these actions a little bit (In some variations, penalize recent actions more)

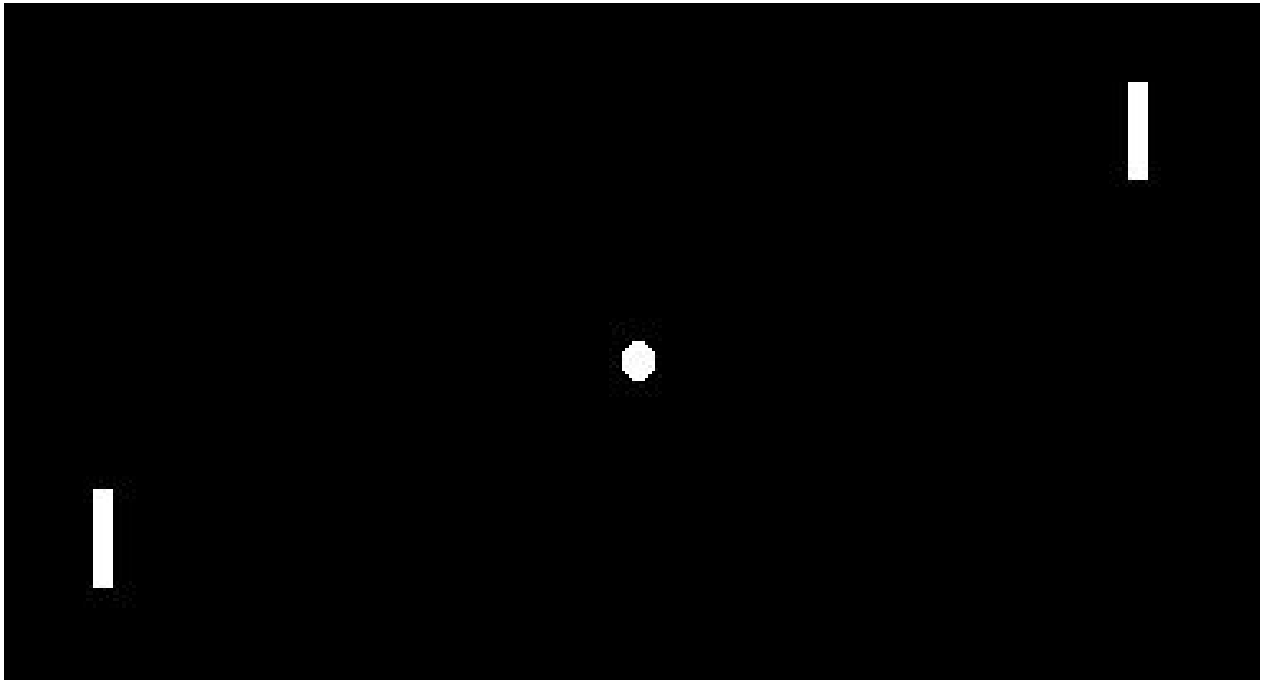
4. Every time we succeed, boost all the actions a little bit

5. Over time, “good actions at the right timesteps” will be boosted, and incorrect actions will be penalized!

Reinforcement Learning

Now make your computer play a few million games of Pong.

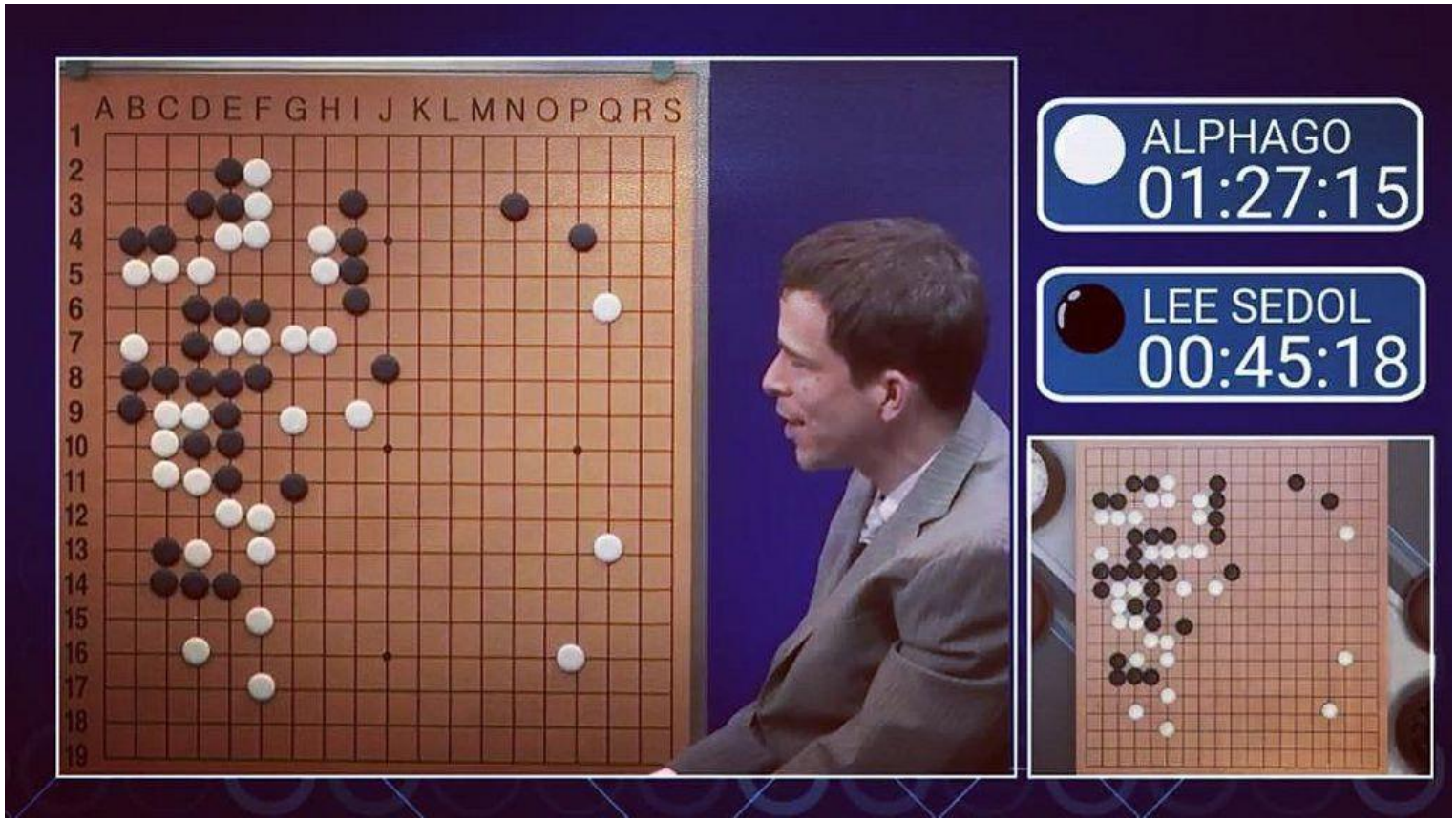
Overtime, it will learn to perform the right actions depending on the location of the ball!



http://karpathy.github.io/2016/05/31/rl/

Reinforcement Learning

More examples: AlphaGo



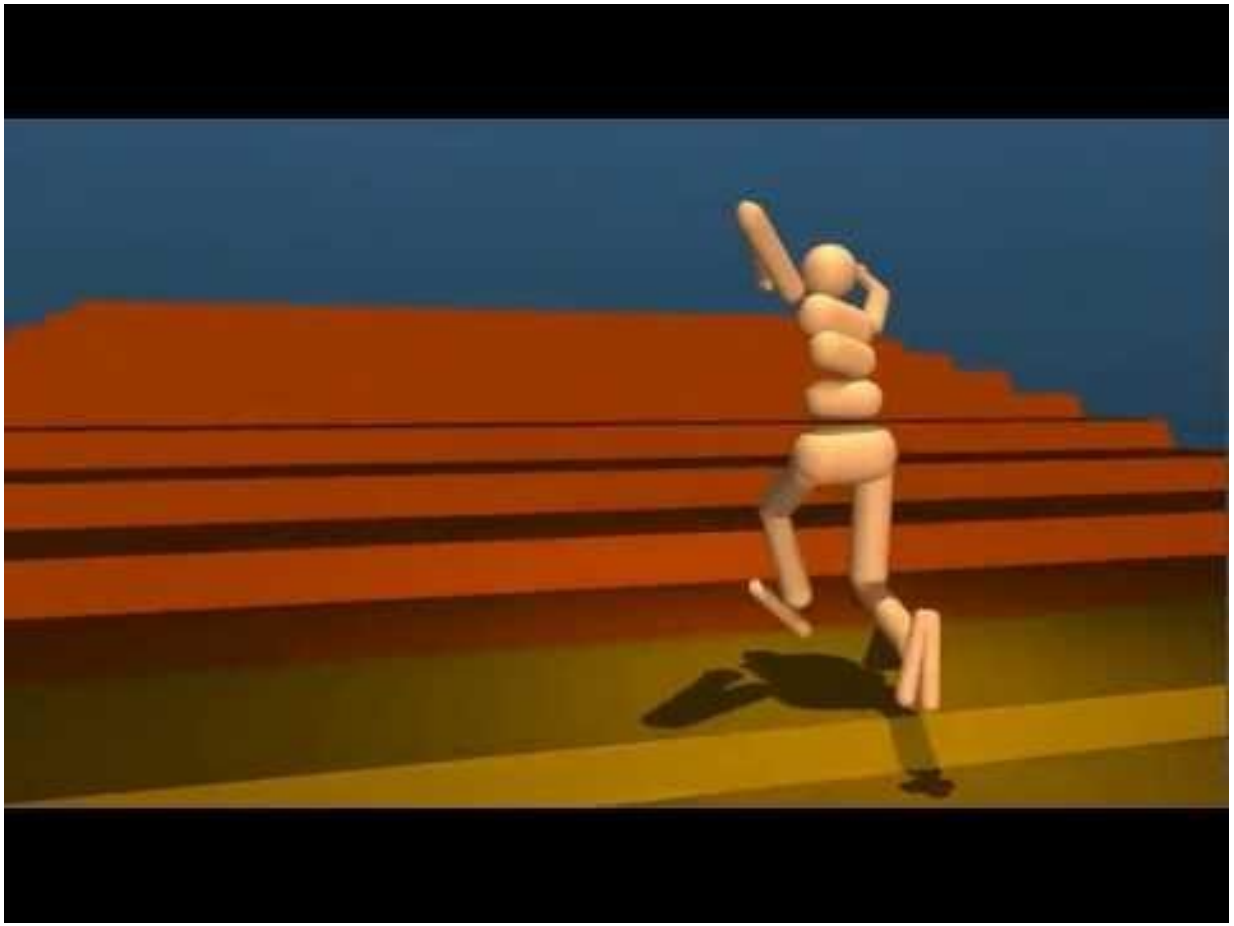
Reinforcement Learning

More examples: DOTA

Reinforcement Learning

More examples: Not just games...

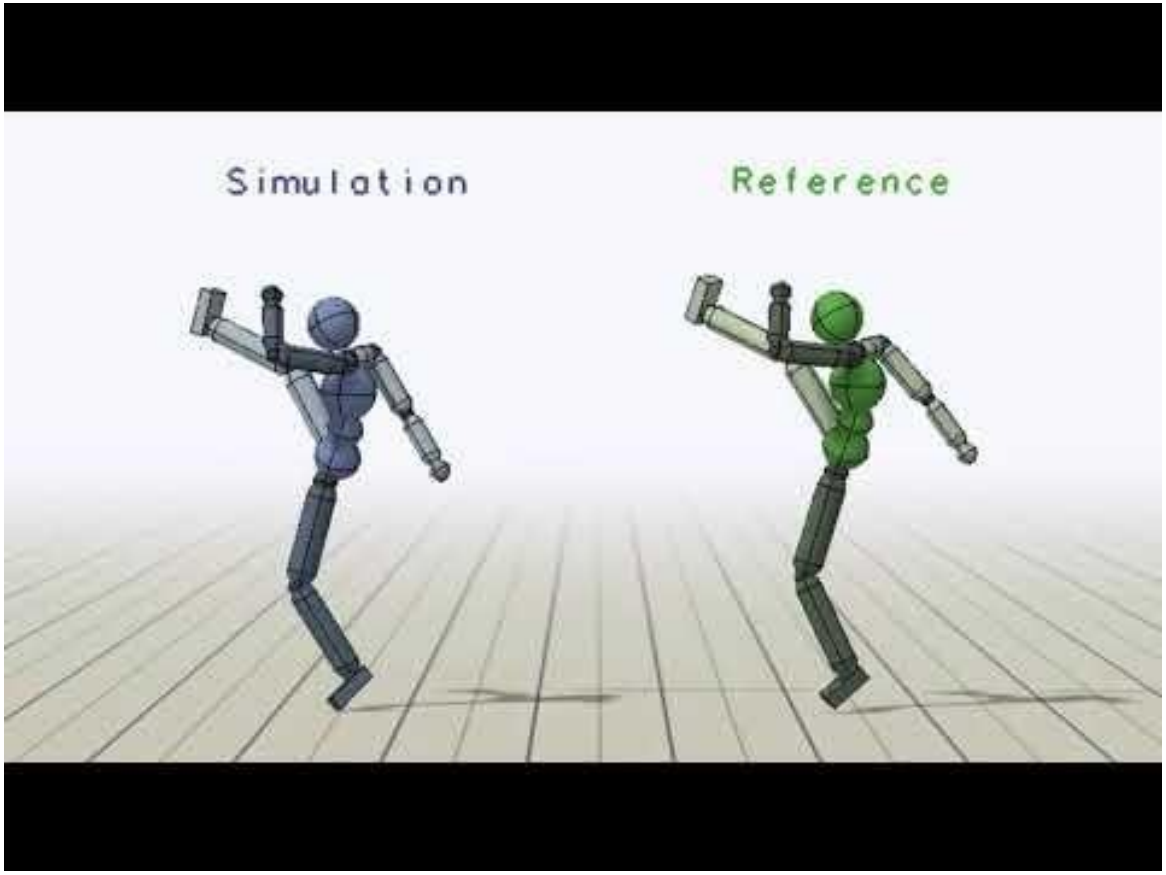
https://www.youtube.com/watch?v=faDKMMwOS2Q



Reinforcement Learning

More examples: Not just games...

https://www.youtube.com/watch?v=XCLSkFKTWyg



Summary

- Domain adaptation aims to use all available data in favor of the in-domain data
- Multilingual systems enable Zero-shot translation
- Multi-task learning improves generalization capability of the model
- Multi-modal learning is the way forward to build general AI's that understand the world like humans do
- Generative Adversarial Networks aim to solve the inverse problem of generating instead of just classifying
- Reinforcement Learning is a super-general framework that can help us teach agents how to act and react in the real world